1 **Monday, 17 March 2025**
2 **(10.30 am)**
3 **LADY HALLETT:** Mr Stoate.
4 **MR STOATE:** Thank you, my Lady. The first witness this
5 morning is Mr Chris Stirling. May he please be sworn.
6 **MR CHRISTOPHER STIRLING (sworn)**
7 **Questions from COUNSEL TO THE INQUIRY**
8 **MR STOATE:** Thank you.
9 Good morning, Mr Stirling, could you please give the
10 court your full name.
11 **A.** Christopher Andrew Stirling.
12 **Q.** Thank you.
13 You very helpfully provided the Inquiry with
14 a witness statement, the reference for which is
15 INQ000561670. It's 41 pages long and dated
16 25 January 2025. Is that statement true to the best of
17 your knowledge and belief?
18 **A.** It is.
19 **Q.** Thank you very much. Just by way of professional
20 background, and before we get to your roles and
21 responsibilities during the pandemic, you say that by
22 profession you're a project manager and programme
23 manager; is that right?
24 **A.** It is.
25 **Q.** Over the last 15 years you have led some major



1 government programmes?
2 **A.** Yes.
3 **Q.** Prior to the pandemic, that included in roles at
4 director level at the Ministry of Justice and, through
5 2019, at the Department of Health and Social Care?
6 **A.** That's correct.
7 **Q.** Two of your previous roles have what you call
8 connections to the broader preparedness agenda, and
9 those are that in 2014, you supported Public Health
10 England around the National Pandemic Flu Service?
11 **A.** Correct.
12 **Q.** And in 2015/16, you led the design of the future and now
13 current operating model for NHS Supply Chain which
14 included things like supply chain resilience and
15 inventory management; is that correct?
16 **A.** It is.
17 **Q.** During the pandemic, you played two, you say, related
18 and overlapping roles which were of relevance to this,
19 the procurement module, and they are, firstly, that from
20 March 2020 to September 2021 you were programme director
21 of the joint DHSC and NHS England Covid oxygen,
22 ventilation, device and clinical consumables response;
23 is that correct?
24 **A.** That's correct.
25 **Q.** And from April 2021 to October 2022, you were interim



1 director of Medical Technology for DHSC?
2 **A.** That's correct.
3 **Q.** So very relevant to the parts of the scope of this
4 Inquiry I'm going to be asking you about.
5 Just to complete the picture, is it correct that you
6 no longer work at DHSC?
7 **A.** That's correct.
8 **Q.** Thank you very much.
9 Mr Stirling, in your very helpful witness statement,
10 you provide, and I'm just going to look at it very
11 briefly, a flavour of the procurement landscape of
12 medical devices like ventilators prior to the pandemic.
13 You say at the time the Medicines and Healthcare
14 products Regulatory Agency, MHRA, had approximately
15 2 million pieces of medical technology licensed for use
16 in the UK market, of which, you say, it was crudely
17 estimated that around 100,000 are in regular use within
18 the NHS every year; is that right?
19 **A.** That's correct.
20 **Q.** And as we have already heard, in the pre-pandemic
21 environment it was the role of NHS Supply Chain to
22 support trusts to source those items effectively; is
23 that right?
24 **A.** That's correct.
25 **Q.** Moving into the pandemic, again, just to give us



1 a flavour of what happened, you say that at the start of
2 the pandemic, issues started to be experienced with
3 suppliers being unable to fulfil demand and trusts being
4 unable to source the required volumes of those pieces of
5 kit through the usual channels; is that right?
6 **A.** That's correct.
7 **Q.** You say that led to early, small-scale, examples of
8 trusts panic-buying, hoarding, and of instances of
9 competition between trusts which drove up prices; is
10 that right?
11 **A.** That's correct.
12 **Q.** You say that while those were initially small, there was
13 an expectation that things would get significantly worse
14 as stock positions and the global trade landscape, as
15 you put it, deteriorated; is that right?
16 **A.** Yes.
17 **Q.** Now, in clear terms in your statement you say that it
18 appeared that the existing NHS structures, both locally
19 and nationally, would not be able to cope effectively,
20 and that an enhanced national-level intervention was
21 required; is that right?
22 **A.** Correct.
23 **Q.** What was the response to that, then?
24 **A.** The response from that from the department was to create
25 number of parallel streams of activity of which the

1 Oxygen, Ventilation, Consumables Response that I led was
2 one of those streams.
3 **Q.** Thank you. I think that was on 3 March 2020 that became
4 active; is that right?
5 **A.** Correct.
6 **Q.** And the scope of that programme, in terms of the key
7 products, we're talking, aren't we, about ventilators
8 and associated capital equipment, clinical consumables
9 and oxygen; is that correct?
10 **A.** That's correct. Initially, the scope was purely
11 ventilators but that grew over time to encompass all the
12 things you have just mentioned.
13 **Q.** Thank you. I want to ask you first, please, in relation
14 to that programme, the programme that you led, some
15 questions about how targets were set and about how the
16 scale of the requirements of these key products was
17 understood. First, in your statement, you say that it
18 was not the responsibility of the programme you led to
19 actually estimate likely Covid inpatient numbers; is
20 that right?
21 **A.** That's correct.
22 **Q.** And so what was it that your programme was responsible
23 for estimating?
24 **A.** So my programme was responsible for taking the demand
25 estimate of Covid cases and converting that effectively



1 to a bill of materials that would be required to enable
2 an intensive care unit to operate.
3 **Q.** What you say in your statement about that is that there
4 was significant uncertainty around trying to get to that
5 number and that bill of materials; is that right?
6 **A.** There is. I think there was significant uncertainty
7 both around what the number was, and therefore, what the
8 bill of materials would be, but also in the early days,
9 significant uncertainty about the treatment patterns and
10 pathways that would be most effective and therefore what
11 materials were required for each patient regardless of
12 the number of cases.
13 **Q.** If I may say, Mr Stirling, some of the figures you give
14 in your statement are quite stark in terms of the
15 requirements estimates.
16 **A.** Mm.
17 **Q.** You say that between 12 February and 17 March 2020, the
18 requirements estimates vary between 59,000, 90,000, and
19 138,000 ventilator beds. On paper, that's quite
20 a significant variation, isn't it?
21 **A.** Indeed it is.
22 **Q.** Now we've already heard in evidence -- the Inquiry has
23 heard from Sir Gareth Rhys Williams that there was, at
24 the outset of the pandemic, no central list of how many
25 ventilators were held by the NHS, what model, spec,



1 whether they were functioning, things like that. That's
2 right, is it?
3 **A.** That's correct. And one of the very first things that
4 NHS England launched prior to my involvement was
5 a survey of the ventilator estate which returned the
6 results on the 6 March.
7 **Q.** Sir Gareth told us in his evidence that there were
8 tentative estimates by DHSC and NHS England that the NHS
9 could have access to something like 6,000 to 8,000
10 ventilators, but looking at the numbers that were being
11 communicated to you, clearly, again, demand for
12 ventilated beds and the associated kit was looking
13 likely to significantly outstrip supply?
14 **A.** Yes.
15 **Q.** I want to ask you about this, because something you
16 specifically note in your witness statement, and I'm
17 just going to read paragraph 39 and then ask you
18 a question about it, if I may. And that's about these
19 requirements estimates. You say:
20 "Whilst they have been subsequently published,
21 distribution of official estimates at the time was
22 highly restricted, even within the health system.
23 Whilst the conclusions of these analyses were shared,
24 typically verbally, the supporting information and
25 analysis was not. This coupled with the uncertainty



1 surrounding speed of pandemic progression, and efficacy
2 of potential interventions created a highly uncertain
3 target-setting environment."
4 What can you tell us about the reasons for the
5 restrictions on estimates?
6 **A.** I believe the restriction on estimates were primarily
7 designed to reduce circulation and avoid some of those
8 estimates reaching the public domain and causing panic.
9 **Q.** What might be surprising, Mr Stirling, is that they were
10 restricted even to somebody in your position, you're
11 doing your --
12 **A.** Yes.
13 **Q.** -- best at that very early stage to get from estimates
14 to, as you say, bill of materials. Why were they even
15 restricted to somebody in your position; do you know?
16 **A.** I think people were very happy to share the conclusions
17 of those activities, and that was sufficient to enable
18 me in my role to pursue all avenues available to
19 increase the ventilation capacity in the country. But
20 I think the desire was to reduce circulation, even
21 amongst individuals such as myself, so that the
22 prospects of any communication would be centrally
23 managed.
24 **LADY HALLETT:** I'm sorry to interrupt, I'm not following.
25 What do you mean by restriction on estimates?

1 **A.** So I was shared the conclusion of the numbers, so the
2 numbers that you've just read out people would tell me
3 "We estimate that they are 59, 79, 138". I was not able
4 to see the documents that led to those conclusions. And
5 so I was working from the headline conclusions rather
6 than seeing and having access to the supporting
7 information that had created them.
8 **LADY HALLETT:** So the rationale was restricted?
9 **A.** So the --
10 **LADY HALLETT:** Distribution of the rationale was restricted?
11 **A.** The logic and the modelling that had created those
12 estimates was not available.
13 **MR STOATE:** Right. And to be clear, we're talking here
14 about estimates of the number of ventilated beds so your
15 job, how are we going to get from what we've got --
16 **A.** Yes.
17 **Q.** -- to what we might need. I just wanted to check with
18 you whether there was any sort of hindrance to your work
19 upon these restrictions on estimates. I will --
20 (overspeaking -- understand that the panic is and
21 restriction -- (overspeaking) --
22 **A.** Surprisingly, I don't believe that that impeded our
23 response. Because the numbers were all so large that
24 they required you to do -- take every and all means
25 available to increase that capacity. And so I think, if



1 the numbers were of a scale where you are considering
2 whether you need to do certain things or not, then
3 having access to the logic and the evidence may have
4 helped us make better decisions.
5 As it happens, all of those targets were so far in
6 excess of the capacity that we had at the time, that
7 I think I can honestly say it didn't influence our
8 decision making as to whether we were going to pursue
9 one route or another route. We were clearly needed to
10 pursue all available routes.
11 **Q.** Just looking to the future, do you think that fewer
12 restrictions on not just the data but the analysis, the
13 modelling, to somebody in your position, looking to
14 scale up medical technology at speed, would it be useful
15 to try to reduce those restrictions or do you think it's
16 not a major issue?
17 **A.** I can't honestly say that that was a major issue in this
18 sense. As a generally inquisitive person, I would have
19 loved to have seen those figures at the time but I can't
20 honestly say I would have done anything differently if
21 I'd have had them.
22 **Q.** Looking at the procurement of ventilators in particular,
23 you do note in your statement that the word
24 "ventilators" does cover or is often used or misused in
25 the coverage of a lot of different types of treatment.



1 We are hearing from Professor Moonesinghe later today
2 who is probably better placed to cover that, but just
3 for context, we're talking about a range of oxygen
4 therapy treatments to support Covid patients, aren't we?
5 **A.** Yes.
6 **Q.** In terms of the purchasing environment pre-pandemic,
7 what you tell us is that before we got to the pandemic,
8 getting a new intensive care ventilator was typically,
9 is this a fair characterisation, a fairly long and slow
10 process?
11 **A.** Absolutely.
12 **Q.** Deliberately and by design?
13 **A.** Indeed.
14 **Q.** Can you set out the main steps.
15 **A.** Typically a trust would identify that they had a need,
16 either because they were creating new intensive care
17 capacity or existing devices were reaching the end of
18 their life. Clinicians would usually engage in
19 a reasonably lengthy testing and assessment process,
20 normally involving getting physical test devices from
21 suppliers, forming a set of testing and evaluation
22 criteria. A specification would be developed,
23 a procurement process would then ensue. Once that had
24 completed, a contract would be let, and a reasonably
25 extensive training, familiarisation and onboarding



1 process would then take place. For a trust adopting
2 a new make or model of ventilator, this process could
3 easily take a year or 18 months.
4 **Q.** It hardly needs saying, I suspect, that you didn't have
5 that kind of time when it came to the start of the
6 pandemic. How would you describe the changes to that
7 procurement process amid, as you say, "a compressed
8 timescale, amid a global stampede of demand". What were
9 the key changes that you had to ...
10 **A.** I think there were changes that we made and there were
11 also changes that the market and the environment made
12 for us.
13 **Q.** Yes.
14 **A.** I think the predominant change we had to make was we
15 were buying at a distance. We were often buying devices
16 that we had never physically held in our hands before,
17 and we were buying from specification.
18 We were also initially engaging with a large range
19 of intermediaries who had come into the market at that
20 time, and the nature of the market demand was such that
21 this was often a cash-up-front make a decision within
22 hours, otherwise someone else will buy the device.
23 So the normal process of challenging suppliers and
24 checking "Does your device do this? Can it do that?
25 How does it respond in this situation?" was not

1 available to us, and we had to largely buy off-paper
2 specifications at a distance.
3 Q. So you've gone from literally months to hours,
4 effectively, in some cases?
5 A. And in some cases, minutes.
6 Q. One other change, we've already touched upon NHS Supply
7 Chain procurement. You note in your statement that it
8 was NHS Supply Chain that held a number of standard
9 framework agreements for the provision of ventilators.
10 I was interested by the aspect of the timeframe on that.
11 You say this:
12 "From programme inception on 3 March, requests for
13 availability went to suppliers on 4 March, first
14 contracts were placed on 9 March and this route was
15 largely exhausted by 12 March."
16 So in a matter of days, the normal NHS Supply Chain
17 route for the provision of these ventilators was just --
18 had evaporated?
19 A. I think that should also be taken as a success of NHS
20 Supply Chain.
21 Q. All right. Why do you say that?
22 A. They provided, I think, close to 40% of the devices that
23 we ended up using. All of those devices had previously
24 gone through extensive testing and due diligence and had
25 commercial value-for-money benchmarks on them, so we



1 weren't paying Covid premiums for those devices. And
2 they were able to very quickly deliver. So you could
3 say it would be lovely if they had been able to deliver
4 all of the requirements, but we were able to get,
5 I think it was, 38/40% of our intensive care ventilators
6 that we ended up purchasing through that route in a very
7 secure way very quickly. So I think those dates and
8 that speed is an example of the supply chain procurement
9 process working really well. The problem was that it
10 couldn't give us everything that we needed.
11 Q. It couldn't meet the demand?
12 A. Couldn't meet the demand.
13 Q. One of the strategic issues -- in your quite reflective
14 statement you talk about some of the lessons you've
15 learned about strategic issues facing medical
16 technology, and you say that one of your key lessons
17 learned from the pandemic experience is that:
18 "NHS Supply Chains needs to either be given a clear
19 mandate (and associated funding) to act as 'supplier of
20 last resort' or not. At present, it is not commissioned
21 to do this, but should issues arise, is expected to do
22 these."
23 In the context of answers you've given there, could
24 you just expand on why you say that and what your lesson
25 learned is about NHS Supply Chain here?



1 A. Well, NHS Supply Chain is designed effectively as an
2 opt-in service that trusts can choose to use or can
3 choose to do different things if they wish. There is no
4 part of NHS Supply Chain which is designed or scaled to
5 act as a provider of last resort.
6 Q. And just to be clear, what is it you mean by that?
7 A. So when the -- when NHS trusts go to NHS Supply Chain,
8 they go and say, "I would like you to get this for me."
9 And in the relationship between trusts and NHS Supply
10 Chain, there is no commitment or no guarantee from NHS
11 Supply Chain that they can supply those items.
12 NHS Supply Chain is set up as effectively a best
13 endeavours service: "We will try to get you those items
14 if we can." It's not set up as a service where we are
15 guaranteeing, as NHS Supply Chain, to provide those
16 items to you.
17 They could provide that function, but they're not
18 currently financed, scaled or resourced to provide that
19 function. The problem was, in a pandemic, all of the
20 NHS trusts suddenly turn to NHS Supply Chain and say,
21 "Please give me these things", and NHS Supply Chain
22 demand suddenly goes through the roof and they're not
23 scaled, planned, expecting to deal with that scale of
24 response.
25 Q. Thank you very much.



1 One of the workstreams we've heard about and that
2 you alluded to earlier was the call to arms and we've
3 already heard from Sir Gareth Rhys Williams about that.
4 You do -- just in relation to that, you do make the
5 observation, in your paragraph 66, that you recall
6 saying to Steve Oldfield at DHSC that you:
7 "... saw no reasonable prospect of achieving the
8 demand projection using the standard projection/NHS
9 Supply Chain means, and that to materially improve our
10 current projection, we would need significant additional
11 support from across government."
12 Is that right?
13 A. That's correct.
14 Q. So we heard about the make drive, we'll be discussing
15 later with another witness some of the work of the FCDO.
16 A. **(Witness nodded)**
17 Q. Foreign and Commonwealth Office, which was mobilised to
18 support the purchase of ventilators from overseas. You
19 also note that the Ministry of Defence was tasked to
20 offer additional support on procurement, logistics and
21 transportation; is all that right?
22 A. That's correct.
23 Q. You make this point: that on 25 March, your department
24 and NHSE issued guidance to trusts setting out the scope
25 of the national procurement effort and in effect asking

1 them not to buy ventilators directly; is that right?
2 A. That's correct.
3 Q. Why was that done?
4 A. Partly it was done to try to avoid trusts competing with
5 each other and with us to procure ventilators. At the
6 time, intermediaries and suppliers who had devices were
7 very keen to offer them to as many different people as
8 possible, in order to drive the price upwards, and we
9 were actively discouraging throughout the entirety of
10 March trusts from seeking to engage directly with
11 suppliers and intermediaries.
12 So I think the guidance that was issued on 25 March
13 really formalised the position that had in fact been in
14 effect from significantly earlier, if not from the start
15 of that month.
16 Q. In terms of offers that came through UK-based suppliers,
17 you say, your paragraph 75, that you and your programme
18 were:
19 "... given a clear steer from ministers in line with
20 agreement from [the Treasury], that the primary
21 consideration was ensuring capacity to save lives over
22 value for money."
23 Is that right?
24 A. That's correct.
25 Q. So the Inquiry has already heard from the chief



1 secretary to the Treasury at the time, Mr Barclay, about
2 how that applied to PPE; possibly, for obvious reasons,
3 that clearly applied to your programme too, did it?
4 A. Yes.
5 Q. You make this observation:
6 "In this context, decision making was dominated by
7 confidence and delivery risk, rather than price
8 considerations."
9 Could you just give us a slightly more simple
10 explanation of what that means?
11 A. I think that means we accepted that we were going to be
12 paying over the pre-pandemic price for all goods, and
13 typically, we often paid up to double the pre-Covid
14 price for items. That is not to say that price did not
15 have any impact upon our decision making, and there were
16 many items that we were offered that were at, you know,
17 10, 20, even 50 times the pre-pandemic price, which we
18 chose not to pursue.
19 Q. So if and when the Inquiry looks at the question of what
20 was spent, value for money, those types of key
21 procurement questions --
22 A. Yeah.
23 Q. -- there was some measure of control exercised, was
24 there?
25 A. Indeed, and I think, you know, the average price of the



1 ventilator that I think we purchased through this
2 initiative was £22,000 for an intensive care ventilator.
3 The pre-pandemic norm for that was around £15,000. So
4 we did pay a Covid premium, but we didn't pay exorbitant
5 or extortionate amounts of money to secure stock at any
6 price.
7 Q. Thank you. I'd like to ask you some questions about
8 clinical due diligence, if I may. Now, appreciating
9 you're not a clinician, you're very clear about that,
10 but you'll understand that it was obviously an important
11 part of the procurement process, wasn't it, for --
12 A. Indeed.
13 Q. -- a machine like a ventilator that has such obvious
14 implications in terms of clinical safety? In terms of
15 clinical due diligence you say that once machines and
16 equipment came in, is this right, they came to
17 MoD Donnington where they were quarantined and it was
18 a team of clinicians who undertook a clinical due
19 diligence process?
20 A. Yes, there were two stages to the final process.
21 Q. Right.
22 A. For all new items that had not previously been used in
23 the NHS, they would arrive at MoD Donnington, they would
24 be unpacked, and there was a clinical due diligence team
25 with clinicians, and they would unbox a sample of the



1 devices. They would check for obvious damage, and also
2 check that all of the necessary components were there,
3 and identify any issues with the devices that they could
4 tell from a visual inspection. Things like, you know,
5 we need an adaptor for this thing, we need a different
6 type of oxygen sensor, it doesn't come with an
7 expiratory block, we need to include one of those. We
8 would then seek to rectify those issues and the devices
9 would then be sent to either MD-TEC in Birmingham.
10 Q. With Professor Clutton-Brock?
11 A. Indeed. Or to Nottingham with Dan Clark. They would
12 test the devices in a lab environment, you know,
13 connecting them up to sensors, simulation equipment, to
14 assess their performance, and they would put the devices
15 in one of four distinct categories. And that was the
16 end process. That process was not in place on day one.
17 That process was in effect in there from about, I think,
18 week three or so. So one or two devices did arrive
19 before that process had been set up --
20 Q. Right.
21 A. -- which we may come to --
22 Q. I suspect we may come to. But looking at those
23 outcomes, then, at risk of dwelling too much on figures
24 here, I would like to know, there are -- you say there
25 were four different outcomes. So once you've -- you've

1 unpacked it, you've remedied the immediate issues, and
2 then there are, is this right, four --
3 A. So the four outcomes are after the second --
4 (overspeaking) --
5 Q. -- understood. Outcome 1: de-quarantine the device and
6 release them into the allocation process without further
7 guidance. Is that right?
8 A. Correct.
9 Q. That's the hope, presumably, that's the best outcome?
10 A. Yes, although, frankly, these are very complicated
11 devices.
12 Q. Of course.
13 A. And very few of those devices would require no further
14 guidance whatsoever.
15 Q. Understood, although you say that accounted for four of
16 the new models of ventilators purchased and assessed by
17 your programme -- yeah?
18 A. Correct.
19 Q. Outcome 2: de-quarantine and release into the allocation
20 process, as you say, accompanied by some guidance,
21 a guidance note, about what you call "necessary
22 workarounds to assist in safe operations". What does
23 that mean, please?
24 A. That means that we were able to deploy them, but there
25 was a set of guidance notes that went out. Now, those



1 guidance notes could have been as simple as, I remember
2 for one device, the power button was in an unusual place
3 on the back of the device, and the guidance note
4 literally was "The power button is where you don't
5 expect it".
6 Q. Right.
7 A. You know, it's of something that's of no real
8 consequence. And indeed, some of those ranged to more
9 complex feedbacks saying, you know, "This setting works
10 like this rather than like this, and if you're going to
11 do this, you probably need to do that as well."
12 Q. Right.
13 A. And some of that is more complicated, some of that is
14 more simple, but these are, you know, green lighted to
15 go out into the system as a result of that testing
16 process.
17 Q. With a note attached?
18 A. With a note attached.
19 Q. Again, that's another four of the models purchased
20 through the programme; is that right?
21 A. Yes.
22 Q. Outcome 3:
23 "Hold the device in quarantine pending supply of
24 additional components, communications, or resolutions to
25 issues identified."



1 That might be obvious. "It looks okay, but we
2 haven't got the right pipes," or something like that.
3 A. It could be that, or we haven't got the right oxygen
4 sensor, or actually, the training manual needs these
5 things to be clarified, and there is work to do before
6 we can safely distribute and deploy these devices.
7 Q. That was six models?
8 A. Yes.
9 Q. Finally:
10 "Do not release the device into the allocation
11 process".
12 A. Yes.
13 Q. Reasons of safety, presumably, clinical concern?
14 A. Or that there were potentially issues that we didn't
15 believe we could address in a reasonably quick or easy
16 timeframe.
17 Q. In time, understood. And that was two?
18 A. Yes.
19 Q. Two models.
20 A. Yes.
21 Q. Importantly, you say in terms of the spend on these
22 models, outcomes 1 and 2, they were what you were hoping
23 for, presumably?
24 A. Yes.
25 Q. That covered 92.5% of the device spend, did it?



1 A. Correct.
2 Q. Outcome 3 was 3.7%, and outcome 4 was 3.4%.
3 Significantly smaller figures. What, Mr Stirling, did
4 the figures, in terms of the categories that these new
5 devices ended up in, tell just, in your view, about the
6 overall value of -- value for money of the programme?
7 A. I think in terms of our ability to secure appropriate
8 quality advices, I think it shows that we overcame the
9 challenge of trying to buy at distance reasonably well.
10 You know, we were buying off a specification off a piece
11 of paper, and when the devices turned up, and we then
12 put them through a test process --
13 Q. Yes.
14 A. -- some of them didn't do what we wanted them to do, but
15 the majority of the money we spent was on devices that
16 did what we were expecting and hoping they would do.
17 Q. That were safe and useable within the NHS context?
18 A. And were deployed into and then used by the NHS.
19 Q. You note that as part of the initial surge response, as
20 you call it, trusts were requesting, again, perhaps for
21 obvious reasons, that your programme send them
22 ventilators as quickly as possible.
23 A. Yes.
24 Q. You say at that point speed of arrival was their
25 overwhelming priority in order to build capacity ahead

1 of expected demand.
2 Yes?
3 A. Yeah.
4 Q. And in line with that priority, your initial operating
5 process was to deliver to trusts without undertaking any
6 additional testing or checking process steps. Is that
7 what you were saying, is that that three-week period
8 before those --
9 A. Agreed, yes. So in the very early weeks, the pressure
10 from trusts was to deliver devices to them directly, and
11 in that period, we had not yet set up the testing
12 processes --
13 Q. Understood.
14 A. -- and systems. So for a very small number of devices
15 that arrived in those first weeks, they were delivered
16 to trusts without the testing and checking process that
17 we'd just gone through.
18 Q. You say that for devices that trusts were familiar with,
19 generally that was fine, I think, is the --
20 A. Yes.
21 Q. -- the tenor of your statement. Clinical and
22 engineering teams understood them, had previous
23 experience, and they could deal with any anticipated
24 challenges.
25 But you say that for devices that were new to trusts



1 or, indeed, new to the NHS that approach created
2 a variety of issues --
3 A. Yes.
4 Q. -- from the relatively straightforward but time
5 consuming, things like the wrong plug or the wrong
6 connector, through to what you call "deeper concerns
7 about device quality and safety"; is that right?
8 A. And I think there I am principally referring to the
9 Shangrila devices, which I'm sure you will come to.
10 Q. We will. You'll understand that the context for these
11 questions, Mr Stirling, is that the Inquiry has received
12 a number of statements from across the NHS and
13 representative bodies, some of which, you will have
14 seen, do --
15 A. Yeah.
16 Q. -- raise concerns.
17 I want to look first, please, at INQ000536367.
18 This is the witness statement of Ashley Shaw, the
19 medical director of Cambridge University Hospitals NHS
20 Foundation Trust. Paragraph 28. I'm just going to read
21 it to you if I may, Mr Stirling.
22 "Adult critical care ventilators were provided by
23 the government which were a completely new model to the
24 Trust. While initial support for the installation and
25 training of these devices was provided there did not



1 appear to be long-term plans in place in relation to
2 ongoing training, maintenance, spare parts and
3 consumables provision. Many teams had to take the
4 minimal training materials and train themselves in the
5 specific nuances of the devices, based on previous
6 experiences. In addition, many of the devices and
7 consumables came from China, and the instructions for
8 use and associated documentation was accordingly in
9 Chinese. On occasion, DHSC/NHSE endeavoured to obtain
10 translations of the documentation, but these varied in
11 quality."
12 Before we get to any specific devices, do you
13 recognise those types of issues arising? Did that of
14 feedback reach you and your team?
15 A. I absolutely recognise the issues, and initially we did
16 not have long-term support and maintenance plans in
17 place. Over time, they were put in place, as you would
18 expect, but the initial focus was very much on securing
19 the capacity and then providing those support
20 arrangements subsequently.
21 There is a reference here to "minimal training
22 materials". I think that is very fair. I think we need
23 to put that in context though, in that normally the
24 training process for a new ventilator device would be
25 very extensive and that in the time and pressure that



1 trusts were under, clearly we were not able to undertake
2 that degree of training and familiarisation that we all
3 would have liked. I think the training materials
4 themselves improved over time, largely due to feedback
5 from trusts.
6 So I very much recognise these issues. I think we
7 worked to try and address them, but certainly on
8 day one, a lot of these things were not in place, as
9 you -- well.
10 I think this statement -- this is Cambridge, isn't?
11 Q. Yes.
12 A. So they predominantly took a device which had
13 a long-time support relationship from Ortus, and so
14 those things appeared but were not there on day one.
15 Q. Before we turn to any specific devices, I want to give
16 you a chance to set the context, because in your
17 statement you say as follows -- it's your paragraph 122
18 if you need the reference:
19 "It is important to recognise that ventilators are
20 highly complicated technology devices requiring
21 regulator support and maintenance."
22 That was the focus of that paragraph there, wasn't
23 it, from Dr Shaw?
24 A. Mm-hm.
25 Q. You say this:

1 "As such, it is not unusual for devices to encounter
2 problems and there were, and continue to be, a regular
3 stream of upgrades, safety notices and refinements to
4 instructions etc, regardless of product quality.
5 Similarly, product recalls are not uncommon, and
6 typically reflect clinicians' and suppliers' desire to
7 reduce potential risks, however small they ... be."
8 Yes?
9 **A.** Yeah.
10 **Q.** So we will bear that in mind when we look at these
11 devices. You're effectively saying people are
12 hypervigilant about safety in such complex advices and
13 in the context in which they're used; is that --
14 **A.** That's correct. And also, I think -- I think it's
15 important to recognise that many of the safety notices
16 that are published are examples of suppliers being
17 diligent, and making sure that they are flagging
18 potential risks. I'm also aware of some perception that
19 if you have a high number of safety notices related to
20 a device, that that device is therefore unsafe. It is
21 often, actually, on the contrary: what that is showing
22 is that the supplier is being extremely diligent in
23 publishing and communicating any potential issues about
24 devices.
25 **Q.** All right. Well, we'll take that as background context



1 when we look at the device you mentioned earlier known
2 as the Shangrila 510S. Yes?
3 **A.** Yes.
4 **Q.** I want to turn up, please, first, INQ000520831.
5 This is the witness statement from another trust,
6 Mr Stirling. This is Toby Lewis, chief executive at the
7 time, of the Sandwell and West Birmingham Hospitals NHS
8 Trust.
9 Paragraph 6:
10 "Our need for ventilators had been duly registered
11 in the submissions made to NHS Midlands. Our projected
12 figures were wholly consistent over time, and consistent
13 with our agreed plans. Until the time specified in the
14 following paragraph [which we'll come to], I had no
15 forewarning of who, how, or when decisions about
16 allocation or specification would be made. This made it
17 difficult to prepare teams of staff for receipt or
18 manage their understandable anxiety about our future
19 readiness."
20 In respect of ventilators, yes?
21 Just pausing there, one of the things that you note
22 in your statement and one of the key parts of your
23 programme was communications; is that right?
24 **A.** Correct.
25 **Q.** You say:



1 "A key aspect to the programme was communicating
2 positions to Trusts both directly and through regional
3 networks."
4 Yeah?
5 **A.** **(Witness nodded)**
6 **Q.** This here, I don't want to be unfair but it looks like
7 an incident, doesn't it -- an instance where
8 communication with a particular trust was less effective
9 than it might have been, to put it neutrally?
10 **A.** Agreed.
11 **Q.** Was that something that you were able to improve over
12 time, overall communication with trusts about what
13 they're getting, when they're getting it, who made the
14 decision and why?
15 **A.** I think the allocation process worked predominantly with
16 regional stakeholders, and regional stakeholders managed
17 the majority of the communications to individual trusts.
18 Our team would deal with individual trusts around
19 specific delivery arrangements and specific issues and
20 challenges, but the majority of the request allocation
21 process was dealt with at a regional level.
22 We published on a daily basis the lists of devices
23 and the inbound schedule of devices that we were to
24 receive, and trusts would engage with their regional
25 teams about their requirements and put in, effectively,



1 requests for devices from those.
2 Now clearly in this example that communication has
3 not worked as effectively as it should. I perceive that
4 generally it worked better than this particular example
5 would demonstrate. I think it also got better over
6 time, and there were a number of significant
7 communication activities to representatives from all
8 trusts saying, "This is how we are working, this is what
9 we are doing."
10 **Q.** All right.
11 **A.** I think there is -- also, the last point here is I think
12 probably, in my view, the most important one: we were,
13 through this point, not having sufficient stock to meet
14 all of the requests that we had in. And I think
15 somewhere else in this statement the author refers to it
16 as a just-in-time model. We would love to have had
17 sufficient quantities of devices to not operate
18 a just-in-time model, but unfortunately the reality of
19 our stock position and the number of demands that were
20 coming in, is that we weren't able to do that.
21 **Q.** Moving on to the specific advices, we can see at
22 paragraph 7 on the screen:
23 "On 6 April, 2020, we received ventilators, which we
24 had been alerted to on [the 3rd]."
25 "8. We received 30 Shangrila 510 ventilators ...

1 directly sourced by DHSC/NHSE ... from the Chinese
2 supplier ..."
3 And delivery came, as you said, through
4 MoD Donnington.
5 The point made in the next paragraph is that:
6 "The scale of [that] delivery made to the Trust was
7 25% less than we had asked for and required for our
8 approved plan."
9 Again, what's said there is that it was unclear how
10 this decision had been made or by whom. I think you may
11 have covered that in your last answer.
12 **A.** I think we frequently delivered people less than they
13 asked for, but we believed we always delivered them more
14 than they needed. And I think later on in this
15 statement that point is echoed about the trust having
16 sufficient ventilation capacity.
17 In terms of how the decision has been made, we
18 operated a process of daily device allocation panels
19 which were chaired by Professor Keith Willett, and had
20 regional representation, typically regional medical
21 directors or their delegates representing the requests
22 and needs of their region.
23 **Q.** If we could go over to paragraph 11, please.
24 "On receipt [this is of this particular device], our
25 engineering and clinical team sought to assess the



1 functionality and capability of the equipment received.
2 Limited information was available to work with.
3 Initially the Shangrila ventilators were unusable as
4 supplied, as the oxygen piping/connectors were not UK
5 standard, the oxygen fuel cells were unusable without
6 modification, and the supplied breathing circuits were
7 multi-use but had no sterilisation instructions. There
8 was no UK representation of the parent company and no
9 one we could obtain advice for consumables from. We
10 then invested significant time modifying the ventilators
11 so we could even test them in our engineering workshop,
12 let alone use them on a patient. We also invested
13 significant time trying to construct a disposable
14 breathing circuit that could safely be used by utilising
15 available components from assorted UK suppliers."
16 And over at paragraph 14, please.
17 "The delivery of ventilators we received came with
18 just two days of consumables necessary to use such
19 ventilators and no information about future consumable
20 supply. This, in addition to the underlying issues
21 above, meant that we were unable to plan to use the
22 ventilators to the extended extent. Any use of the
23 machines was exploratory and under close senior clinical
24 supervision over the latter part of the week commencing
25 April 5 ..."



1 Clearly, Mr Stirling, quite significant concerns
2 from the clinician's --
3 **A.** Indeed.
4 **Q.** -- point of view. What is it you want to say about this
5 particular device, why it ended up where it did and what
6 was done to address those concerns?
7 **A.** So this was the very first device that arrived.
8 **Q.** Right.
9 **A.** And it arrived prior to the establishment of the testing
10 processes and, as we talked about before, was delivered
11 straight to a trust without having gone through that
12 test. It was a device that was from a new supplier, new
13 to the NHS. It was also a transport ventilator which we
14 thought could potentially be used in an intensive care
15 environment.
16 I think it's also important to say that the device
17 itself is extensively used in other parts of the world,
18 particularly in China and the Far East, and is from one
19 of their very reputable ventilator manufacturers.
20 The specification for the device had been given to
21 clinicians to see whether they believed it could fulfil
22 a need, and on the basis of the paper specification, we
23 had decided to procure these devices.
24 These were also very low-cost devices. As
25 I mentioned earlier, I think the average price of an



1 intensive care ventilator we bought was £22,000 a unit.
2 I think these devices were around £3,000 a unit. They
3 were purchased on the basis that we were currently at
4 that point in time projecting to be incredibly short of
5 devices, and looking to see whether devices such as this
6 could fulfil a gap in an emergency.
7 The feedback we got from the West Midlands from
8 their testing evaluation of this device was very
9 disappointing. And I think that was communicated to
10 Professor Moonesinghe, who you're seeing later, on
11 12 April, and the ventilators were withdrawn from the
12 allocation process and re-called on 13 April.
13 **Q.** Yes. And, in fact, if we look, please, at
14 INQ000514215 -- I'll show you that page, that's your
15 name and the date of 13 August 2020?
16 **A.** August, yeah.
17 **Q.** If we go up to the top, please, the first page. We
18 don't need to go into any detail here but we can see
19 this is a note from you called "Shangrila 510s
20 Cancellation Refund Position". So by August you were
21 seeking to cancel the orders, which you describe as now
22 surplus to requirements, and if this recommendation,
23 your recommendation, is that right, Mr Stirling, was
24 accepted, DHSC would expect to receive a refund in the
25 sum of 2.8 million and make a loss of approximately

1 1.2 million on that transaction?
2 A. That's correct. There are some things between then and
3 this letter that probably should be noted.
4 Q. Okay.
5 A. We immediately worked to cancel down future orders.
6 Q. Yeah.
7 A. We then subsequently put this device through the testing
8 process that we discussed earlier.
9 Q. Mm.
10 A. And it is one of those two devices that ended up in
11 outcome 4.
12 Q. Didn't make it.
13 A. The least preferred outcome.
14 Importantly, we also used that process to understand
15 whether there were any discrepancies against the
16 specification, things that we could go back and say to
17 the manufacturer "Your device did not do what it was
18 supposed to do."
19 And importantly, the conclusion from the test lab
20 was that whilst they were equally disappointed with the
21 performance, there was no smoking gun that they could
22 say, "It didn't do this, therefore you can, you know,
23 have a contractual recourse of some nature."
24 Therefore, we engaged with the manufacturer of those
25 devices and sought to negotiate a refund, and I think



1 further down in this note I think it says they offered
2 us I think it was a 70% --
3 Q. Yes, exactly that.
4 A. -- refund.
5 Q. Yes, we don't need to go in detail to the note but you
6 achieved, and one of the documents you provide in your
7 statement suggests that that was in fact received, not
8 long after. So a 70% refund was achieved.
9 Importantly there you say you used -- did you use
10 the learning from that experience to try to --
11 A. I think this, as the first device to arrive, was
12 instrumental in helping us set the processes up.
13 Q. Right.
14 A. To make sure that future devices, as they arrive, were
15 properly tested, no matter what the pressure from NHS
16 trusts to receive devices promptly might have been.
17 Q. Thank you. I want to look very briefly, please, before
18 we break, at issues which arose around another device
19 and how you sought to tackle those and the learning that
20 was taken forward there. This is the UCL Ventura,
21 a continuous positive airway pressure, or CPAP, form of
22 oxygen therapy; is that right?
23 A. That's correct.
24 Q. You tell us that early work for that device had been
25 carried out as part of the Ventilator Challenge, about



1 which we've already heard, but it ultimately didn't come
2 through that process. Can we look, please, at
3 INQ000514162.
4 This -- yes.
5 A. Yes, I recognise this.
6 Q. You recognise this. Very good. This is your submission
7 for the purchase of these devices, is that right, in
8 treating Covid-19 patients?
9 A. That's correct.
10 Q. Could we go to page 4, please. Can you see under this,
11 I wanted to briefly ask you about this -- let's set the
12 context. There's a CPAP device, you're recommending in
13 this document that they are purchased; is that broadly
14 it?
15 A. That's correct.
16 Q. Under the heading "Publicity" at paragraph 22 here:
17 "The Chancellor of the Duchy of Lancaster has
18 already announced (31 March) that the NHS has already
19 ordered a quantity of these devices. This was in line
20 with the ongoing contractual discussions at the time."
21 Can I ask you, did the fact that that had already
22 been announced in any way affect your decision to
23 recommend the purchase? This wasn't, was it,
24 procurement by press release in some way?
25 A. No. We had already -- the contractual paperwork,



1 including this formalisation submission, took a while to
2 complete, as these things do, and so we had previously
3 discussed and engaged in discussions with UCL around our
4 intention to buy these devices. So I think this is
5 a press release getting slightly ahead of factual, and
6 I think if he had announced that we intended to, or that
7 we were planning to, or working with UCL to, then he
8 would have been absolutely spot on. At that point we
9 had not technically ordered the devices or placed the
10 contract to do so.
11 Q. I want to know whether this created any form of
12 ministerial pressure that led to some of the later
13 issues?
14 A. I don't think this created any pressure or influenced
15 the decision in any way, shape or form. It was
16 something that we were committed to working towards, and
17 this is the submission that formalises that.
18 Q. Thank you. That's very clear.
19 If we just look at paragraphs 19 to 21 then briefly,
20 this is under the heading over the page "Risks", is
21 what's headed above there.
22 Some risks were noted, weren't they, in relation to
23 this device, which you've clearly set out?
24 A. Agreed, although these are not risks that are
25 exclusively associated with this device.

1 **Q.** Absolutely.
2 **A.** Risks with all CPAP devices. And indeed, one of the key
3 benefits of this device was that it used less oxygen
4 than some other CPAP devices, even though in relative
5 terms, that is still quite a lot of oxygen.
6 **Q.** You say:
7 "Whilst oxygen systems have alarms designed to
8 prevent this happening ..."
9 So trusts' systems failing, we can see in 19.
10 **A.** Yeah.
11 **Q.** "... even relatively short interruptions in supply could
12 have significant patient safety impact."
13 Again, for quite obvious reasons.
14 "To mitigate this, extensive work is being done with
15 Trusts to ensure that they understand, and work within,
16 their oxygen flow constraints. Devices will be
17 appropriately labelled and contain significant warnings
18 to ensure that they are only used in the appropriate
19 situations."
20 So that was what you noted. Unfortunately, what we
21 can see from one trust is that that labelling and
22 warning became one of the issues with this device,
23 didn't it?
24 **A.** I think the labelling was always more restrictive than
25 the usage authorisation.



1 **Q.** I see.
2 **A.** So there were three labels at different times that
3 should have been on these devices.
4 **Q.** Can I pause you there, Mr Stirling, if I may?
5 **A.** Yeah.
6 **Q.** Let me put the context for that question and then give
7 you an opportunity to answer it in a way that you may
8 have just been about to.
9 The context for this question is another trust
10 statement, I think the last I'll refer you to. This is
11 the Royal Surrey NHS Foundation Trust.
12 Lawrence, please INQ000512921, paragraph 6.6.
13 It's the statement of Louise Stead and Jacqui Tingle
14 of that trust.
15 "... [DHSC] provided us with ..."
16 And then a number of different figures are given
17 about different devices. We don't need to go through
18 all them but there's ten of a GE device, 80 Flo-Ox
19 oxygen monitors, 26 VisionAire 5 portable concentrators.
20 And in the middle there, we see 80 of these UCL Ventura
21 CPAPs that you and I are discussing.
22 What Ms Stead and Ms Tingle say is that:
23 "Other than the Visionaire 5 Portable Oxygen
24 Concentrators, these devices were not used, primarily
25 due [to] safety concerns of non-standardised devices,



1 lack of appropriate consumables, and no competency
2 training for clinical teams. For example, the UCL
3 ventilators had labels advising they should not be used
4 for clinical use."
5 That's quite the restrictive label.
6 **A.** I'm not sure that is the complete text of the label.
7 **Q.** All right.
8 **A.** So my recollection is that there were three labels. The
9 first label said, "For test and evaluation purposes
10 only", which was the most restrictive of the conditions,
11 and that was to enable the devices to be used
12 specifically in line with MHRA testing requirements.
13 The second label said, "Should not be used for
14 clinical use outside of the pandemic."
15 And I think this may have chopped the last few words
16 of that only.
17 **Q.** All right.
18 **A.** Because MHRA exceptional use authorisation, once
19 granted, had been given for Covid-19 patients only.
20 And the third label was a general "You can now use
21 this for anything" label in line with the standard
22 things.
23 The normal process for re-labelling devices is that
24 you recall all of the devices, get them sent back to you
25 physically, you change the label on the device yourself,



1 and then you send the device back out.
2 Given that in many cases these devices were being
3 used for patient care, that was clearly inappropriate to
4 do that, and it was agreed with MHRA that we should have
5 a modified process where we sent the labels out to the
6 trusts for them to over-label the devices.
7 Clearly in a pandemic environment, re-labelling the
8 devices was unlikely to be the trust's priority action,
9 and I'm aware that many of those devices were not
10 re-labelled and so effectively had, potentially, an
11 earlier label rather than an older label -- I said that
12 wrong.
13 **Q.** The next witness from whom we're hearing, or one of the
14 next witnesses, I should say, as you're aware, is
15 Professor Ramani Moonesinghe.
16 I want to bring up one paragraph of her statement,
17 please.
18 INQ000518349, page 31, it's paragraph 115.
19 And just to conclude this section of questions
20 before, my Lady, if we may take our break,
21 Professor Moonesinghe will tell us, I'm sure, that she
22 had quite a significant role, didn't she --
23 **A.** She did.
24 **Q.** -- in deciding which bits of kit and in particular which
25 ventilators and associated consumables were procured and

1 weren't. She says this:
2 "Although the target number of machines was
3 achieved, the quality of the machines and their
4 suitability for use in the NHS, to treat Covid-19 or
5 other conditions was variable. There were also
6 shortages of other essential items, including some
7 medicines and consumable items. These limitations meant
8 that clinicians were not always able to treat patients
9 with their first choice of interventions."
10 Now, on clinical choice, you might be happy,
11 everyone might be happy to hear is my suggestion is that
12 we deal with that after we have had a break because
13 I suspect there is a bit to say, Mr Stirling, but on the
14 quality of what was procured, Professor Moonesinghe's
15 view is it's variable and that certainly seems to be the
16 flavour of the statements I've put to you.
17 **A.** Mm.
18 **Q.** Do you agree with that conclusion?
19 **A.** I think it clearly was variable. The simple reality
20 was, we would have loved to have only procured the
21 highest quality equipment all of the time, but we were
22 simply not able to achieve the targets by doing that, so
23 we had to look for alternative sources, and I think if
24 you go back to the Cambridge statement from earlier,
25 they ended up with a mixed fleet of Servo devices that



1 they had originally, and devices that we had procured
2 for them. I don't think there's any doubt they would
3 have much preferred to have had an entire fleet of their
4 first choice Servo devices. I would have, if I was in
5 their position. However, we were not in an ability to
6 supply that, so we had to supply other devices that were
7 not necessarily as fully functional, and didn't have all
8 of the features that their first choice device would
9 have done. I think that's reasonably self-evident.
10 **Q.** Do you think this variability in quality was a result of
11 having to procure in such large numbers at such speed?
12 **A.** Yes. Clearly if we'd have had, you know, ten years to
13 build up the numbers that we needed, we would have been
14 able to operate differently and secure more first choice
15 devices for trusts.
16 **MR STOATE:** My Lady, just before we turn to the next topic,
17 that might be a convenient moment.
18 **LADY HALLETT:** Certainly. You were probably warned that we
19 take a break but I promise you we will finish your
20 evidence before lunch.
21 11.45, please.
22 **(11.29 am)**
23 **(A short break)**
24 **(11.45 am)**
25 **LADY HALLETT:** Mr Stoate.



1 **MR STOATE:** Thank you, my Lady.
2 Mr Stirling, I'd like to turn now, please, to some
3 questions possibly foreshadowed by the last answers you
4 were giving about allocation of goods and the question
5 of clinical choice, in other words, did the clinicians
6 get a choice? To what extent did they get that choice,
7 and how was that challenge addressed?
8 You say in your statement that an allocation process
9 was developed to ensure that the devices we've been
10 discussing were appropriately and fairly distributed.
11 Yes?
12 **A.** Correct.
13 **Q.** Could you talk us through that -- what process was
14 developed?
15 **A.** So the process worked as follows: we would distribute to
16 trusts via regional teams a list of the ventilation
17 devices and equipment that we had available, together
18 with a projection of our incoming stock, so they could
19 see what was coming down the line.
20 **Q.** What you had and what was coming?
21 **A.** What we had and what was coming.
22 **Q.** Yeah.
23 **A.** Yeah. And obviously in the pandemic environment we
24 didn't always get what was coming exactly right but at
25 least we had an idea of what was coming.



1 Trusts were then invited to put in requests for
2 equipment and those requests were -- arrived at us
3 through the regional teams, and so the requests that we
4 ended up considering were ones that had come from
5 trusts, had the support of the regional teams. And
6 those were requests for individual devices and device
7 types, because they could see the makes and models of
8 things.
9 And obviously, some of the devices we had were
10 incredibly popular, and some of the devices that we had
11 were less popular.
12 We operated two different models, one for during
13 surge conditions where we distributed devices on the
14 basis of urgent clinical need. And in those situations
15 we would look at a trust's ventilation and intensive
16 care capacity, we would look at the number of patients
17 they had within that and what degree of free capacity
18 they had and the rate of change of that, and on the
19 basis of that, we would say, "Yes, you have an urgent
20 need for additional capacity", in which case we would
21 prioritise that request.
22 Outside of surge times, we operated
23 a capacity-building model where we would seek to help
24 trusts build and correct capacity for the next surge.
25 In terms of the devices, we would try and make sure

1 that wherever possible, trusts were using familiar
2 devices or using a small number of different devices.
3 Clearly, if we are going to give people unfamiliar
4 devices, it's better for us to give them 20 of the same
5 unfamiliar device, rather than 20 different devices.
6 And we would also prioritise those organisations that
7 had small needs and small capacity over those that had
8 very large needs.
9 So, for example, the Cambridge example we were using
10 earlier, I think they ended up, from that statement,
11 with 67 VG70s that we supplied.
12 **Q.** That's a model of ventilator?
13 **A.** That's a model of ventilator that we sourced as part of
14 the new, novel sourcing process.
15 We had a very small number of Servo devices, which
16 I believe would have been their preferred make and
17 model, but we would have delivered them to an
18 organisation that maybe only needed two or three extra
19 ventilators, in which case we were able to keep them on
20 their same device type, whereas with Cambridge they were
21 going to have to use two different device types because
22 there was no way we were going to be able to supply them
23 with 70 additional Servo devices. So we would try and
24 prioritise the models to reduce variation from people's
25 initial device choices.



1 However, we weren't always able to do that and there
2 were situations where we had to give people more
3 variability in devices than they would have liked.
4 **Q.** You talk in your statement about the national
5 ventilation allocation panel.
6 **A.** Yes.
7 **Q.** First of all, what was that and how did it fit into this
8 process you're talking about, please?
9 **A.** So the national ventilation allocation panel was the
10 decision-making group that was comprised of regional
11 representatives and chaired by Keith Willett or one of
12 his deputies, that made the decisions about which
13 devices were sent where. So, effectively, the
14 culmination of the process that I've described of we
15 provide lists to trusts of what makes and models we've
16 got coming in when. They put requests in. Those are
17 considered by the regional teams and then at that
18 national ventilation allocation panel, typically
19 3 o'clock, seven days a week, the decisions would be
20 taken on which devices would be sent to which trusts.
21 Those would typically be communicated to the
22 warehouse by 4.00, and the devices would be delivered
23 either that evening or, typically, early the following
24 morning.
25 **Q.** We'll come back to the question, in a moment, please,



1 about which devices, and who wanted what, in terms of
2 models and makes, and so forth. In terms of the
3 question of how many, was that also a question for the
4 national ventilation allocation panel? "We've got this
5 many, we can see your need, you can have X"?
6 **A.** Exactly.
7 **Q.** I want to put to you, please, a part of the statement of
8 Professor Banfield, chair of the British Medical
9 Association, the BMA's UK council.
10 It's INQ000562457, paragraph 262.
11 I just invite your views, if I may, please.
12 Professor Banfield tells us:
13 "Intensive care provision was stretched across the
14 UK during the pandemic, despite capacity being increased
15 in a variety of ways, including using spare and new
16 ventilator machines, altering staffing patterns,
17 redeploying doctors from other departments and expanding
18 into other clinical areas beyond the normal ITU
19 footprint such as theatre recovery areas."
20 And then this:
21 "There were, at times, localised shortages of
22 ventilators which necessitated the transfer of patients
23 between hospitals, and the use of anaesthetic machines
24 in lieu of ventilators, and the use of anaesthetic
25 machines ..."



1 Forgive me, that's the end of it. I think it might
2 be duplicated in the sentence there. But the point is
3 he's saying there were at times, in the BMA's view,
4 localised shortage of ventilators --
5 **A.** Yes.
6 **Q.** -- which necessitated transfer between hospitals.
7 And then if we look down, please, at paragraph 264,
8 "Impact on members":
9 "Shortages of ventilators for the numbers of
10 patients who might benefit, combined with a lack of
11 government or NHS guidance on clinical prioritisation,
12 put additional pressure and uncertainty on doctors
13 during an already highly stressful period.
14 Circumstances in which demand outstrips supply raise
15 serious ethical and professional challenges and give
16 rise to the potential for moral injury among those
17 doctors responsible for making final decisions on care
18 escalation, as well as wider concerns about the
19 potential for discrimination in the application of
20 criteria."
21 Focusing, then, on what's said here, because I'm
22 asking you again about the 'how many', localised
23 shortages of ventilators which necessitated transfers
24 and caused this impact on BMA members. Firstly, do you
25 recognise that a concern which arose?

1 **A.** Could we go back to 262.
2 **Q.** Yes, of course.
3 **A.** The previous one. That would be very helpful. Thank
4 you. So I think I've got a slightly different view on
5 this. A critical care transfer of a ventilated patient
6 is a considerable and difficult undertaking, and not
7 something the system would do lightly.
8 It's comparably significantly easier to move
9 a ventilator to where it's needed. So I don't dispute
10 that there were transfers of patients between hospitals
11 to places where there was greater clinical care,
12 critical care capacity. I am not aware of any situation
13 where the reason for that transfer was that there was no
14 ventilator.
15 I think it's very clear that at times in different
16 part of the system, critical care was extremely
17 challenged. What I'm not aware of is any time where
18 there was critical care capacity -- there's workforce,
19 oxygen, medicines, beds -- all set up but no ventilator.
20 To the best of my knowledge, that did not happen.
21 So I think there's a bit of a sort of switching
22 between shortage of ventilators and shortage of critical
23 care capacity here.
24 **Q.** For which ventilators is a vital part?
25 **A.** It's a component --



1 **Q.** A component?
2 **A.** -- but a part of --
3 **Q.** Exactly, but one component of --
4 **A.** -- that broader picture.
5 **Q.** All right.
6 **A.** So I think, you know, I think that -- the use of
7 anaesthetic machines in lieu of ventilators I think was
8 relatively common at the time, and clearly not optimal.
9 I'm not the person to talk to the clinical
10 appropriateness of that or not.
11 **Q.** No, indeed.
12 **A.** But I'm aware that that was part of what was quite
13 widespread at the time. So I think that -- agree with
14 that bit of it. And clearly, on 264, there is no doubt
15 that not having a -- not having the machine that you are
16 familiar with, and lots of it available, will increase
17 stress and pressure in the clinical community.
18 So these points about, you know, the fact that you
19 haven't got your ideal first choice and you haven't got
20 confidence that you've got a lot of it, clearly that's
21 going to create pressure.
22 **Q.** Yes.
23 **A.** But I do disagree with the core premise that there were
24 localised shortages. If there were, I'm not aware of
25 it, and that's not something that we've seen evidence



1 previously.
2 **Q.** Thank you. Let's look, then, at the question of
3 clinical choice. Just to set the background, you make
4 a number of points in your witness statement about
5 clinician choice. First, paragraph 12, you call it
6 a "key, structural, element of the UK health system";
7 yeah?
8 **A.** Agreed.
9 **Q.** Paragraph 57 you say:
10 "Within intensive care environments, clinicians
11 generally have strong preferences about the ventilators
12 they use, and a reluctance to change. These preferences
13 typically arise from the impact that familiarity and
14 consistency have on reducing patient risk."
15 So, in other words, an understandable preference?
16 **A.** Agreed.
17 **Q.** And at paragraph 116 you say this about the process
18 we've just been talking about:
19 "In determining what makes and models of devices to
20 provide [to trusts], attempts [you say] would be made to
21 provide consistent device types, reducing variation and
22 complexity for clinicians."
23 So not always possible, I think that was the
24 conclusion of what you said about that; is that right?
25 **A.** Yes.



1 **Q.** I want to look briefly, please, at the position that
2 we've been told about in Scotland.
3 **A.** Okay.
4 **Q.** Can I bring up INQ000498141.
5 This is the witness statement, Mr Stirling, of
6 Caroline Lamb, chief executive of NHS Scotland and
7 Director General of Health and Social Care.
8 I'm going to look at paragraph 314.
9 Ms Lamb here is talking here about the Scottish
10 Government ICU Resilience and Support Group, which in
11 Scotland led coordination and decisions on procurement
12 and distribution of ventilators. And she says this:
13 "For ICU ventilators and equipment, the focus was
14 equipping up to the maximum surge Level 3 ICU beds to
15 deliver double the capacity (406 beds) up to the maximum
16 surge Level 3 ICU beds (714 beds), caveated by the
17 ability to staff ICU beds."
18 And then this:
19 "The approach adopted by the ICU Resilience and
20 Support Group [in Scotland] from the start was to take
21 a human factors approach -- ie purchasing and using only
22 familiar brands or brands that NHS boards were using
23 already where possible. This meant that staff working
24 under extreme pressure were not also having to learn to
25 work with new specialised equipment and enabled the

1 equipment to be utilised by Boards post-Covid."
2 Does that in any way differ from the approach that
3 was taken in England, Mr Stirling? Does that sound
4 like, in any way, a different approach?
5 **A.** I'm not sure it does sound different. I think the
6 keywords I would use -- look at, here, are "already
7 where possible".
8 **Q.** Yes.
9 **A.** So clearly I think, both in Scotland and more broadly in
10 the UK, the preference was to use familiar brands and
11 models that people are familiar with. However, the
12 challenge was, I think either in Scotland or in the rest
13 of the UK, whether we were going to achieve those
14 high-volume targets with those familiar brands.
15 I don't think anyone was trying to introduce new
16 things because we thought that was a good thing to do.
17 We thought that was a necessary thing to do.
18 **Q.** One of the reasons I've been asking you these questions,
19 Mr Stirling, is because of an observation you make in
20 the reflections and "Lessons Learned" section of your
21 witness statement. This is your paragraph 203(c), you
22 say this:
23 "The pandemic has highlighted the cost of
24 choice ..."
25 I'll just put what it says here and then ask you to



1 expand on it, if I may. You say this:
2 "... the importance, advantages, and disadvantages
3 of choice have long been debated. The pandemic
4 experience has shown that clinicians can, even under
5 pressure, successfully switch products. It has also
6 highlighted the significant cost and complexity of
7 operating the current 'choice max' [as you call it]
8 environment. Clearly, clinical choice needs to remain
9 central to both clinical practice and innovation, but
10 overall [you say] the pandemic has materially weakened
11 the case for choice."
12 What do you mean by that? Can you just expand on
13 why it is you say the pandemic has weakened the case for
14 choice by clinicians in terms of these devices?
15 **A.** I think, when looking at device selection, there's
16 a number of factors that influence that. And,
17 historically, the NHS system has always prioritised
18 clinical choice above almost all other factors.
19 Resilience implications, industrial strategy
20 considerations, are typically not part of that broader
21 clinician choice consideration. And I think the
22 pandemic has highlighted the quite considerable costs,
23 both financially and in terms of performance, of having
24 such choice embedded at a fundamental level within the
25 system.



1 Clearly, I'm not advocating that we should move from
2 an environment where we have, you know, choice max, as
3 I describe it, to one where you say everybody must use
4 this product, because clearly that has resilience
5 implications of its own. But I think, if there is this,
6 you know, this spectrum of choice from, you know,
7 direction here through to maximum choice over here, the
8 NHS is here [indicating].
9 And I think there is an opportunity for us, in our
10 long-term procurement supply of these devices to move
11 towards a little bit more where we can have structured,
12 long-term partnerships with a smaller number of
13 suppliers that enables us to have longer-term
14 resilience, and also enables us to respond more
15 effectively to situations like this, and to simpler
16 things like, you know, staff moving between hospitals,
17 finding they have to familiarise themselves with
18 completely different sets of equipment. And that has
19 quality and safety implications of its own.
20 So I'm by no means advocating a more draconian
21 approach, but I think we have to be honest that the
22 degree of choice which is embedded within the NHS system
23 has a cost, and the pandemic has shown us, both in terms
24 of quality and pound signs, what that is.
25 **Q.** Thank you. Just to conclude this area of questions,



1 I want to show you a document, again drafted by you.
2 It's INQ000541370, titled --
3 **A.** Ah --
4 **Q.** You know this one?
5 **A.** Yes, know this one.
6 **Q.** It's always pleasing when you know, a document,
7 Mr Stirling.
8 **A.** It is.
9 **Q.** That's my aim. This is one -- it's got your name on it
10 so it's from you to Steve Oldfield, who we know --
11 **A.** No, this one is to --
12 **Q.** -- (overspeaking) -- forgive me, to the Secretary of
13 State, forgive me. And it's a document titled "Medium
14 term strategy to ensure continued availability of
15 Oxygen, Ventilation and Medical Consumables, including
16 procuring a stockpile of Covid-19 related ICU
17 consumables."
18 If we could just look, please, at page 2,
19 paragraphs 2 to 3.
20 This seems to be your summary of this first part of
21 the work of your programme.
22 "During the first surge of Covid-19, the Oxygen
23 Ventilation, Medical Devices and Clinical Consumables
24 Programme has delivered a wide range of activities ...
25 to ensure [the] availability of Oxygen Therapies and

1 associated critical care treatments within the NHS.
2 "Whilst this work has been successful, in that there
3 has not been any reported shortage of Oxygen or
4 Ventilation adversely impacting patient outcomes, it
5 has, on occasion, been uncomfortably close."
6 It was that phrase I wanted to ask you about,
7 Mr Stirling, if I may. What was uncomfortably close
8 about the work of your programme with regard to what
9 we've been discussing this morning?
10 **A.** I think, you know, both in terms of ventilation
11 equipment, oxygen and consumables, the quantities that
12 we had available, we managed to prioritise and deploy
13 across the system. If Covid cases had been 10% or 20%
14 higher, we might not have been in the stage where we
15 were able to make the first half of that statement and
16 say that we had no reported shortages of those items.
17 It certainly wasn't comfortable. I don't think
18 there is any particular deeper meaning to that statement
19 than saying, you know, we were uncomfortably close, in
20 many cases.
21 **Q.** That's very clear. Thank you.
22 Does that apply as much to the devices we've been
23 talking about as it does to oxygen?
24 **A.** I think the timeline of shortage or maximum constraint
25 was different for different items.



1 **Q.** All right.
2 **A.** So for ventilation devices, our position of greatest
3 shortage was on day one. For consumables, it was
4 probably towards the end of the first wave, and for
5 oxygen, it would have been the peak of the second wave
6 in January 2021.
7 **Q.** That's very clear. Thank you, Mr Stirling.
8 I'm going to ask you then briefly about oxygen.
9 Thank you, Lawrence.
10 A key challenge, you say, in your programme when you
11 started out was that -- this is you paragraph 169,
12 Mr Stirling, to orientate you.
13 **A.** Thank you.
14 **Q.** -- was that there was no central system-wide
15 understanding of how the healthcare oxygen
16 infrastructure worked. It rather mirrors the concerns
17 about not much data/no data on ventilators. You say
18 there was no central system-wide understanding of the
19 oxygen infrastructure; is that right?
20 **A.** That's correct.
21 **Q.** No clear national owner, you say, or centre of
22 expertise, and within trusts systems had often been
23 configured and set up a very long time ago. Yeah?
24 **A.** Agreed.
25 **Q.** And you say also that some of that expertise, such as it



1 was, had declined over time. That was your
2 paragraph 169.
3 This is from your paragraph 25:
4 "The additional capacity built over the summer
5 period [of the pandemic] placed significant extra
6 pressure on Oxygen infrastructure which became a primary
7 constraint. In line with expectations, there was
8 significant usage of the consumable stockpiles, and
9 limited use of residual capacity (excepting oxygen
10 infrastructure). Despite daily oxygen management
11 process and resources being severely tested [your
12 words], at a system level, Oxygen capacity plans held."
13 So can we turn to that, then. What was the nature
14 of the severe test of the oxygen infrastructure system?
15 **A.** So I think at some point in the statement -- I'm
16 struggling to find where -- there is an illustration of
17 that in terms of the number of trusts that were using,
18 you know, 50%, 8%, 100% of their oxygen infrastructure.
19 **Q.** Yes, I think it's your 182, if that --
20 **A.** 182?
21 **Q.** Around there.
22 **A.** Thank you. You know this better than I do ... yes.
23 So in normal times, pre-Covid, a trust would expect
24 to run at 20, 30, 40% oxygen capacity.
25 **Q.** Right. Just explain to us in simple terms what that



1 means. You've got a -- oxygen is produced, isn't it, in
2 very simple terms, between non-clinicians here,
3 Mr Stirling, through something called a vacuum insulated
4 evaporator or VIE?
5 **A.** So not quite.
6 **Q.** Right, help us out.
7 **A.** Oxygen is typically produced in, effectively, a factory
8 called an air separation unit.
9 **Q.** Okay.
10 **A.** Which looks, to all extents, like a factory, makes
11 oxygen. And from there it is shipped in cryogenic
12 tankers, to trusts, and is put, in liquid form, in the
13 vacuum insulated evaporators, which is where you were --
14 **Q.** I got that bit.
15 **A.** That's where you were.
16 **Q.** Very good.
17 **A.** And this effectively a huge tank that sits outside the
18 hospital with very cold liquid oxygen in it. There
19 is then a set of equipment called an evaporator and
20 a control panel that converts the liquid oxygen to
21 gaseous oxygen and pumps it into a series of pipes that
22 run through the hospital to get to the bedside.
23 So, from an oxygen infrastructure perspective, there
24 are a number of things that can go wrong: at your
25 factory you can not make enough; in your distribution,

1 you can not get enough to the tank in the hospital; your
2 tank in the hospital can run out; the equipment that
3 converts the liquid to gaseous cannot operate at
4 sufficient capacity, so you just can't turn it into gas
5 fast enough; or you can have insufficient pipework in
6 the hospital to be able to serve all the different
7 places that you need to deliver it, at any given time.
8 So those are the sort of potential points in
9 failure.
10 Now, the first of those, we managed to mitigate very
11 effectively, and our biggest --
12 **Q.** Pausing there, you mean getting enough oxygen to the
13 tank?
14 **A.** Making enough oxygen, delivering it, getting it to the
15 tanks was -- I don't believe that we have any issues
16 across the pandemic environment with those stages.
17 There's a lot of people worked very hard to make sure
18 that was the case, but I believe that was fine.
19 Where we had the biggest issues were in the
20 technical equipment that converts the liquid to a gas,
21 and particularly in the pipework within trusts.
22 Now, it's relatively easy -- I say "relatively
23 easy" -- you know, again, in peacetime it would still be
24 a 6/9-month project, but we had people doing this in
25 days and weeks -- to upgrade the equipment so that you



1 can make more gaseous oxygen quickly. And our
2 industrial partners became very adept at high-speed
3 upgrades of those components.
4 What is much more difficult is to replace the
5 pipework, typically buried within the walls of the
6 hospitals, some of which were designed a very long time
7 ago, because you've got to get to the pipes and then
8 you've got to make them bigger, typically. And that's
9 very hard to do at the best of times, and even harder to
10 do if you're mid-pandemic.
11 So that's was where the majority of our discomfort
12 comes, and that's why the figures that you have here --
13 **Q.** Yes, can you just set those out for us?
14 So normal capacity you're looking at 20 to 40,
15 you -- (overspeaking) --
16 **A.** 20 to 40.
17 **Q.** Yeah?
18 **A.** Yeah.
19 **Q.** And how did that change over the pandemic and why?
20 **A.** So, as intensive care and also the general oxygen
21 support to Covid patients not in intensive care
22 increased, oxygen usage within trusts increased
23 significantly. And so the rate of conversion of liquid
24 to gaseous oxygen, went -- you know, the figures as per
25 point 182, in January '21 we had 55 trusts: ten over 80



1 and five trusts over a 100%. So those are trusts that
2 are, according to the paperwork, running their system
3 far higher and hotter, or in this case cooler, than was
4 ever intended.
5 Now, thankfully, there is some resilience in those
6 specifications, so people were able to successfully
7 operate oxygen systems at above 100% but that is clearly
8 not sustainable, recommended --
9 **Q.** Or potentially safe?
10 **A.** Or potentially safe. However ... I'll leave it there.
11 **Q.** Okay.
12 Just in terms of the steps taken, then, to bring
13 that down, so we've got a situation by January '21 --
14 **A.** Yeah.
15 **Q.** -- that you've told us about. How did your programme
16 briefly, please, mitigate those effects? In procurement
17 terms, what was done?
18 **A.** So in procurement terms, we worked with oxygen-providing
19 organisations to upgrade components of infrastructure.
20 And NHS England provided a variety of financing to
21 trusts to enable them to remove bottlenecks within their
22 pipework systems. Our team also produced a series of
23 oxygen best practice guidance to help trusts use less
24 oxygen.
25 **Q.** Yeah.



1 **A.** Which was actually surprisingly effective, because prior
2 to the pandemic nobody had really put much effort into
3 optimising oxygen usage.
4 And I think we also worked to swap out higher oxygen
5 flow usage devices for lower oxygen flow usage devices.
6 So these are devices that have the same clinical
7 benefit but simply consume less oxygen to do that. So
8 those were the different sort of strands of activity
9 that took place over summer and autumn of '20 in order
10 to be ready for January '21.
11 **Q.** In terms of the lessons you learned from that work, your
12 paragraph 204, thinking specifically now about oxygen,
13 one of them is, you say, that something called the
14 MGPS HTM needs updating. What is that, please?
15 **A.** That's the Medical Gas Pipeline System Healthcare
16 Technology Memorandum, which is effectively the critical
17 document which defines the dos and don'ts for oxygen
18 systems in hospitals. You know, it is the manual of how
19 to specify and run a hospital oxygen system.
20 **Q.** And why do you say it needs updating?
21 **A.** You know, because it's 20 years old and there have been
22 significant technical advances during that period and
23 also, in particular, there is something within it called
24 the diversity assumptions, and that's flow diversity,
25 and this is a ratio that says how many of the endpoints

1 in a hospital do you have to be able to support at any
2 given time? And that influences the size and quality of
3 pipework.
4 **Q.** Right.
5 **A.** Now, the assumptions pre-pandemic are clearly not
6 appropriate for a post-pandemic environment, and I
7 believe they need to be revised.
8 **Q.** That's very clear. Thank you.
9 Secondly, you make this observation: UK oxygen
10 production will naturally decline over time.
11 **A.** Yes.
12 **Q.** Why is that and what do you say we should do?
13 **A.** So we were relatively lucky in the UK, as a former
14 steel-producing country that used to make large amounts
15 of steel. Steel consumes large amounts of oxygen in its
16 production. So the UK was incredibly well endowed with
17 air separation units, the oxygen factories. And that
18 was a significant boon to this pandemic.
19 In the absence of there being a consistent demand
20 for that oxygen, the UK industrial ability to produce
21 oxygen will naturally decline as plants reach their end
22 of life, and there is no economic rationale for people
23 to replace them. So in 10, 20, 30 years' time, we will
24 have significantly less oxygen capacity than we have
25 today.



1 The government has a variety of things it could do.
2 It can either, you know, take steps to incentivise the
3 retention of that capacity, or seek to invest further
4 upstream, you know, if we had a steel industry in the
5 UK, we would be in no danger of seeing a reduction in
6 our oxygen capacity.
7 **Q.** Thank you. Turning to a different topic again, please.
8 Not much longer to go.
9 I want to ask about work with the devolved
10 administrations. You say the programme worked closely
11 with the devolved administrations.
12 **A.** Yes.
13 **Q.** Some activities undertaken like the procurement of
14 additional ventilation capacity and the one we've been
15 discussing, the boosting UK oxygen production, were done
16 at a UK-wide level. So clearly took in the DAs as well,
17 yes?
18 **A.** Yes.
19 **Q.** You say that the strategic intent was to do as much
20 together as practical and beneficial whilst respecting
21 the devolution of authority.
22 **A.** Correct.
23 **Q.** And you talk about a typical discussion with devolved
24 administrations. What would that look like?
25 **A.** Typically, we would outline our plans to them, and they



1 would decide whether they wanted to participate in those
2 or not.
3 **Q.** In any given part of the programme, or as a whole?
4 **A.** Generally, in specific elements of the programme. We
5 would say, "We are about to do this thing, we can either
6 do it UK wide or we can do it as England. What do you,
7 Scotland, Wales, Northern Ireland, think about that?"
8 And in some situations, all of them participated.
9 In some, at times there was often very good reason for
10 Northern Ireland doing things quite differently, partly
11 because of their connections to Southern Ireland and the
12 way gas operations in particular work over there. But
13 generally, people would be happy to work with us on
14 elements and also do their own things as well.
15 **Q.** One more question, having promised none, on allocation.
16 You say that the devices we've been discussing procured
17 by your programme on behalf of the UK, were offered to
18 the devolved administrations on what you call a crude
19 but widely recognised per capita basis?
20 **A.** Yes.
21 **Q.** And you say, given differing administration needs and
22 equipment profiles, actual devices delivered would often
23 change delivering more of some items and less of others
24 whilst remaining broadly within the intent of that
25 per capita distribution. Yes?



1 **A.** Agreed.
2 **Q.** My question for you is this: as far as you were aware,
3 did any officials or anybody else from the devolved
4 administrations raise concerns with you about that
5 allocation principle?
6 **A.** No.
7 **Q.** I want to turn briefly, then, to some concluding points
8 as lessons learnt. One more bit from
9 Professor Moonesinghe, if I may.
10 INQ000518349. This is her paragraph 187. This is
11 her reflection overall. She says:
12 "I do not consider that there was a robust system in
13 place to ensure an adequate supply of key healthcare
14 equipment and supplies to the NHS during the initial
15 phase of the pandemic. As far as those areas in which
16 I was involved -- specifically equipment, consumables,
17 medicines and oxygen, the efforts made by DHSC, Cabinet
18 Office and NHS England were remarkable and effective in
19 many ways; however, because of a lack of availability of
20 resources to meet the demand in the first wave,
21 clinicians were required to use unfamiliar or less than
22 ideal equipment, medicines and consumables, which will
23 have augmented the pressure under which they were
24 working. These factors combined posed a potential risk
25 to patient safety and may have added additional

1 psychological burden to staff who were working under
2 conditions of unprecedented challenge."
3 I want to focus on that very first sentence, the
4 "robustness of the system in place." what's your
5 reflection on that, Mr Stirling? Do you agree with
6 Professor Moonesinghe?
7 **A.** I think it's very easy to say that we would have been in
8 a stronger position if we'd have had more supplies of
9 the items that we needed on the shelf already. In terms
10 of the ventilators, and the consumables, I think the
11 second point she makes about using items that were
12 unfamiliar or less than ideal. I think that's the key
13 bit that I would focus on and take here, because in many
14 situations it wasn't that we didn't have an item, we did
15 have an item. But it wasn't necessarily the item that
16 people were familiar with or would have considered their
17 first choice.
18 So I think, practically, I think we would struggle
19 to see what a system would look like that would
20 guarantee clinicians their first choice item in a time
21 of unprecedented demand and pandemic.
22 So I certainly think it could have been more robust,
23 there's certainly greater levels of preparedness and
24 stocks that could be held in advance. And that would go
25 some way to easing these pressures. I'm not sure



1 whether it's realistic to have as an objective that
2 everybody always gets their first choice item. I think
3 that's quite -- whether that's an achievable objective
4 I'm not certain.
5 **Q.** Your personal recommendations: first, at paragraph 187,
6 you talk about the requirement for a stronger national
7 medical technology capability. So zooming out from
8 ventilators, putting ourselves in the situation of
9 a future pandemic, which requires scaling up some piece
10 of complex medical technology at pace, what do you mean
11 by -- it might be obvious, but a "stronger medical
12 technology capability"?
13 **A.** So if we compare medical technology to medicines,
14 medicines has had, for an extended period of time,
15 a core national team, both within the Department and
16 within the NHS England, that have, as their enduring
17 remit, making sure the country is in an appropriate
18 place with medicines.
19 Prior to the pandemic, no one had that remit for
20 medical technology, and one of the things that I did in
21 my later stages of my involvement with the Department,
22 was create the Directorate of Medical Technology, which
23 was aiming to provide a central point of leadership for
24 that going forward.
25 I think that has benefits in both a pandemic



1 resilience environment and also more broadly, supporting
2 issues around choice, availability, and innovation.
3 **Q.** You say this:
4 "Medical technology needs clear ownership: it is now
5 too big, and impactful to patient outcomes not to have
6 clear direction and leadership."
7 Is that what you were seeking to address through
8 this directorate?
9 **A.** Yes.
10 **Q.** You're not there any more but was that work under way as
11 you left?
12 **A.** Prior to my leaving, I acted as the interim director of
13 Medical Technology for the Department, and also
14 supported the Department in recruiting a permanent
15 successor, who is still in post.
16 **Q.** Very good.
17 Another of your observations is this:
18 interoperability of medical devices and technology
19 should be mandated.
20 Paragraph 203.
21 **A.** Yes.
22 **Q.** Could you just tell us what you mean by that, please?
23 **A.** So a number of these devices, device manufacturers will
24 often use bespoke connectors, bespoke plugs, slightly
25 different formats and styles of doing things.



1 **Q.** And is that where we can see some of the problems that
2 the clinicians I've put to you -- (overspeaking) --
3 **A.** Absolutely. And if you talk to suppliers, they will say
4 that that is done in the name of patient safety. And
5 there are very important patient safety benefits of
6 making sure that you can't accidentally plug the wrong
7 thing in. That doesn't necessarily mean that every
8 supplier needs to have their own bespoke piece. And
9 there are also obvious economic benefits to you as
10 a supplier if your device only works with one connector
11 and you have to therefore get all of your consumables
12 back from the device.
13 Now, certain suppliers have features which enable
14 you to use their devices in interoperable modes, and my
15 belief is that it should be a requirement for all future
16 medical technology that if there is something that is
17 bespoke or unique, or prevented as part of normal
18 operation, there is a way of turning that feature off in
19 a pandemic environment where you may have to plug it
20 into something else. It's not particularly conceptually
21 difficult, but what it does mean is you don't have to
22 run completely different supply chains for breathing
23 circuits. All of the filters are inoperable: the oxygen
24 sensors, all of the little bits and pieces that actually
25 make this work, you don't have to have ten different

1 types of each of those. I also think that would make it
2 a lot easier for clinicians.
3 But again, it's a change that would need to be
4 brought in in a phased and measured manner.
5 **Q.** Thank you.
6 Finally, please, I want to put one of the author
7 recommendations of Professor John Bell, who was our
8 supply chains expert. It his author recommendation 17.
9 I don't need it to be put up on the screen. What
10 Professor Manners-Bell says is this:
11 "If new health and care equipment is required as
12 part of pandemic response, efforts should be focused on
13 providing existing medical equipment manufacturers with
14 the resources to scale up production rather than
15 encouraging designs from manufacturers with no relevant
16 experience. Production under licence should be
17 a consideration."
18 What's your take, your view on that, please,
19 Mr Stirling?
20 **A.** I understand where he's coming from. My experience of
21 having been through the pandemic is that, actually,
22 existing medical device manufacturers tended to be
23 slower and more risk averse in their willingness to try
24 unusual and challenging things that need to be done,
25 whereas suppliers from other industries are potentially



1 more flexible and willing to operate on the basis of
2 "This is what I can do". We did have discussions with
3 many medical device manufacturers about scaling up their
4 capacity, and probably better taken up with Gareth and
5 the Ventilator Challenge team.
6 **Q.** Yes, we did do so, yeah.
7 **A.** But my experience was that non-healthcare suppliers were
8 faster, more nimbler and agile at responding to our
9 needs than existing medical suppliers. And I was
10 surprised by that. So I can see where he's coming from,
11 but that doesn't tie with my experience.
12 **MR STOATE:** Thank you.
13 My Lady, those are my questions, I think --
14 **LADY HALLETT:** Thank you.
15 Some questions first of all from Ms Morris, who is
16 just sitting there.
17 **Questions from MS MORRIS KC**
18 **MS MORRIS:** Thank you, my Lady.
19 Mr Stirling, good afternoon. I represent the Covid
20 Bereaved Families for Justice UK, and I have two topics,
21 please, to ask you questions about. The first is on the
22 topic of preparedness, particularly in relation to
23 consumables, not ventilators, not devices, but
24 consumables.
25 You say at paragraph 141 of your statement that when



1 we entered into the pandemic -- in a strong position at
2 all levels, there's -- you say ward covers were well
3 stocked, I think is the phrase. The concern in your
4 statement was about being able to sustain that supply --
5 **A.** Yeah.
6 **Q.** -- should there be a protracted surge or further larger
7 surges as part of the pandemic; is that right?
8 **A.** That's right.
9 **Q.** So, from that analysis, I'd anticipate that you'd accept
10 that the stockpile of medicines available at the
11 beginning of the pandemic was not modelled around the
12 provision of a high or protracted surge demand in the
13 event of a pandemic. It was well stocked but couldn't
14 face the surges that the pandemic inevitably brought?
15 **A.** I think I agree with you. Can I just check one thing?
16 I can't comment on medicines. So I can absolutely
17 respond on consumables, but in terms of the medicines
18 stockpile, that's not something I was responsible for.
19 **Q.** Okay. I am going to ask you now about a report from the
20 Swine Flu Critical Care Clinical Group. So, after the
21 2009 swine flu pandemic, the Department of Health had
22 received feedback from clinicians within that group in
23 which they expressed some concerns about the need to
24 stockpile supplies of consumables and medicines in their
25 example. This is a report that's been shared with you



1 as part of the Inquiry's disclosure to you.
2 In that report they said that in a pandemic of
3 greater severity, sustained production, including that
4 from outside of the UK and haulage, would be in
5 jeopardy. So they were sort of foreshadowing
6 beyond 2009 that, in a more severe pandemic, there'd be
7 a problem with the sustained production of consumables
8 and medicines.
9 They said that a sustained surge of critical care
10 capacity would be very difficult to maintain without the
11 building up of stocks in local units.
12 One of the items they mentioned in their report was
13 propofol, an active sedative component of intravenous
14 anaesthesia, so part of critical care provision.
15 So my questions are this, please: is it correct,
16 then, that the Department of Health was well aware that
17 there would be a lack of sustained production and supply
18 of medicines and consumables during a pandemic, and this
19 would predictably affect ICU medicines like propofol?
20 **A.** Okay, so my involvement in this started on 3 March '20,
21 and I was not involved in the pre-pandemic preparedness
22 activities prior to that date. So I appreciate that
23 document was released as part of the evidence process.
24 I had not seen that document prior to that, nor would
25 I have expected to have done.

1 In terms of the core point around the sustained
2 production of consumables, we import the vast majority
3 of our consumables from overseas. Large quantities of
4 that, in fact the majority, comes from the Far East. So
5 I think you are absolutely correct and fair to say that
6 in a pandemic environment, where global trade and travel
7 is disrupted, and that supply of consumables is not able
8 to be brought in on a consistent basis from other parts
9 of the world, the UK would struggle to maintain
10 ICU capacity.
11 **Q.** Thank you.
12 Is it correct, therefore, to say that the only
13 really predictable anyway of ensuring that supply is
14 maintained in a pandemic is through having adequate
15 stocks, through adequate preparedness, having the
16 cupboards well stocked from the outset?
17 **A.** I think stockpiling is always going to be part of the
18 answer. I think, increasingly, if we're looking for
19 a longer-term and more sustainable answer, on or
20 nearshore manufacturing opportunities, particularly in
21 flexible production capabilities, potentially offers
22 a more cost-effective and sustainable route of achieving
23 that.
24 Stockpiles are very effective, but they are
25 incredibly expensive.



1 **Q.** But they are part of the solution, you will accept?
2 **A.** I think it is difficult to see any credible solution
3 where a degree of stockpiling is not part of that.
4 **Q.** Thank you. That's helpful.
5 My second topic is about the operational management
6 of the NHS Supply Chain. In your statement you
7 recognise the NHS Supply Chain lacked the resources to
8 respond to increased demand during the pandemic. Those
9 are your paragraphs 19 and 21.
10 You said in evidence earlier on this morning that in
11 the pandemic all the trusts turned to the NHS Supply
12 Chain and said, "Give me X", and then they weren't
13 expecting necessarily to deal with the scale of the
14 response during the pandemic, the requests from those
15 trusts. However, do you accept that the DHSC had not
16 assessed the supply chain's readiness and capacity to
17 cope in the event of a pandemic and that subsequent
18 demand that that would entail?
19 **A.** Again, I'm sorry, but I can't really talk to the
20 pre-pandemic preparedness point. I think, to your, you
21 know, NHS Supply Chain was operating I think around
22 60-ish per cent of NHS capacity, you know, higher on
23 some lines, lower on other lines. In the Covid
24 environment, we both had an increase in demand, and also
25 those remaining 40% of trusts seeking to source from NHS



1 Supply Chain. So effectively there was a double
2 increase for Supply Chain, and having previously been
3 involved in, you know, the NHS Supply Chain Future
4 Operating Model planning process, that degree of change
5 would normally be ascribed in years, and I think there
6 is no doubt that NHS Supply Chain could have scaled or
7 provided that service. The challenge was doing that
8 from a standing start at very short notice.
9 **Q.** Understood.
10 Final question, please. We've heard from
11 Professor Sanchez-Graells about, sort of, the complex
12 network of contracts and governance that was involved in
13 the supply chain. Is it correct, in your view, that
14 without sort of a clear picture by the DHSC about the
15 governance and operational management of the supply
16 chain, it would have made it more difficult to assess
17 the need and the impact of the demand in the pandemic?
18 **A.** So the supply chains associated with ventilator
19 productions are typically incredibly complex.
20 I think we tried to streamline our engagement with
21 that supply chain by, at various times, taking decisions
22 not to engage with intermediaries and try to engage
23 directly with the factories producing the devices to
24 attempt to shorten that.
25 More data earlier would have been helpful. However,



1 I think we need to just take a slight degree of caution
2 around the availability of data, because, you know,
3 there are hundreds of thousands of products in use in
4 the NHS and we probably needed and used data for maybe
5 2 to 300 of those products. It would be very easy as
6 a knee-jerk reaction to say we need the data for all of
7 the products, and create a very significant reporting
8 burden and challenge.
9 What we were able to do with the consumables was to
10 introduce a system where we were able to add and remove
11 products to the list that we got data for, as and when
12 market conditions enabled us to do that. So capturing
13 data when we needed it, and, you know, I think we just
14 have to be careful not to have the knee-jerk, we need
15 all of the data, all of the time for everything, because
16 otherwise we will need another thousands of people just
17 to manage sets of data that we're not actually going to
18 use.
19 **MS MORRIS:** Understood.
20 Thank you, my Lady.
21 **LADY HALLETT:** Thank you, Ms Morris.
22 Mr Thomas.
23 Mr Thomas is over there.
24 **Questions from PROFESSOR THOMAS KC**
25 **PROFESSOR THOMAS:** Good afternoon, Mr Stirling, can you hear

1 me?
2 A. Loud and clear.
3 Q. I represent FEMHO, that's the Federation of Ethnic
4 Minority Healthcare Organisations, and like Ms Morris,
5 I have two topics that I would like to explore with you.
6 Mr Stirling, given your role, and given the
7 disproportionate impact of the virus on ethnic minority
8 communities, it's certainly important for those who
9 I represent to explore how considerations of equality
10 influenced the procurement and distribution decisions.
11 So with that in mind, my first question is this: in
12 paragraph 17 of your statement, you outline the primary
13 aims of the oxygen, ventilation, medical devices and
14 clinical consumables, including ensuring resources were
15 located "in the right place [according to] clinical need
16 rather than strength of voice or procurement
17 capability."
18 Mr Stirling, how were these principles applied to
19 ensure fair and equitable distribution of medical
20 supplies, particularly to healthcare facilities in
21 regions or communities with high concentrations of
22 ethnic minority populations who might have been
23 disproportionately affected by the virus in healthcare
24 facilities serving high risk, ethnic minority
25 communities? Can you help us with that, please.



1 A. Okay. So our allocation process, as described earlier,
2 was focused on looking at the volume of critical care
3 ventilated patients in any area at that time. And we
4 used that to support decisions around allocating
5 ventilators for urgent clinical need. Between surges,
6 the capacity building activities, we had an explicit
7 objective to level up capacity, and by that, I mean
8 increasing critical care capacity in those areas that
9 had least capacity, more than in those areas that
10 already had higher levels of capacity. And so the
11 intent, through that, was to reduce any discrepancies in
12 capacity levels between regions.
13 We also looked at that through the lens of
14 specification of devices, and discussed with clinicians
15 what, if any, differences needed to be made to the
16 specification to ensure that all parts of the population
17 were served, and that included buying -- the primary
18 area of their feedback around that was that age and
19 weight were considerations, and a number of devices were
20 bought specifically to support the paediatric and
21 neonatal markets.
22 Q. Thank you. You mentioned age and weight were
23 specifically considered. Just piggybacking on that last
24 answer, was the race considered, ethnic minority? Was
25 that considered?



1 A. So we asked our clinicians to help us get the
2 appropriate specifications, and we asked them what
3 differences we needed to make to support the population.
4 The only time that I'm aware that race became an active
5 consideration was around pulse oximeter devices. But
6 from your reaction, I think you're already aware of some
7 of the discussions and discussions around that.
8 Q. Yes, we are.
9 Let me ask you my next question. During these
10 phases, were there any impact assessments conducted to
11 evaluate how the distribution and accessibility of
12 critical medical supplies affected ethnic minority
13 healthcare workers and communities differently? So did
14 you do any impact assessments?
15 A. So I don't believe we did any impact assessments of the
16 distribution of these devices, no.
17 Q. Okay. Let me move on. In paragraphs 27 to 30 of your
18 statement you describe significant challenges related to
19 the use of ventilators that were unfamiliar to staff,
20 and had documentation primarily in Chinese.
21 Question: how were decisions about the procurement
22 and distribution of these devices assessed to ensure
23 compliance with the Public Sector Equality Duty,
24 particularly in ensuring that fair treatment and safety
25 for all healthcare workers.



1 A. So we recognise that some of the instruction manuals and
2 supporting documentation was in other languages. We
3 arranged for translations of those to be produced. We
4 didn't translate them per se; we produced what we called
5 quick-start guides in English. We did not provide any
6 support to individuals that were not able to read
7 English. I'm sorry, I might have misunderstood your
8 question, but we didn't, for example, translate those
9 documents into Welsh or other languages of any kind. We
10 translated them into English.
11 Q. Okay. Can I move on to my second and final topic,
12 please, and this is looking forward, being future
13 looking. As we reflect on the pandemic's challenges and
14 the lessons learned particularly concerning supply
15 chains, resilience and resource allocation, it's
16 imperative to consider how future policies can be
17 designed to be -- more effectively to support ethnic
18 minority healthcare workers to ensure equitable access
19 during a crisis.
20 So that leads me to this question: in paragraphs 201
21 to 204 of your statement, you discuss the lessons
22 learned from the pandemic's challenges, particularly the
23 need for a broader stockpile and improved manufacturing
24 capabilities within the UK.
25 Given these reflections, how do you propose future

1 policies be adjusted to ensure more inclusive support
2 for minority groups, such as minority healthcare
3 workers, who might be disproportionately affected in
4 such crisis?
5 **A.** I think, clearly, if we have a stronger stockpile and
6 more availability of first-choice advices, to use the
7 phrase from earlier, less individuals will be asked to
8 use devices that they are not familiar with, and I think
9 that will benefit all healthcare workers.
10 In terms of the appropriateness of those devices,
11 I think we were, and would always want to be, clinically
12 led by what was appropriate to be procured. And so we
13 were working from the guidance of Professor Moonesinghe,
14 who you're seeing later, and others, as to what would be
15 needed for the entirety of the healthcare worker
16 population.
17 So I think there is clearly things that we can do to
18 make sure we are better prepared, and I think they will
19 benefit all parts of the workforce.
20 **PROFESSOR THOMAS:** My Lady, thank you, those are my
21 questions.
22 **LADY HALLETT:** Thank you very much, Mr Thomas.
23 Those were the questions we had for you,
24 Mr Stirling. You've described your role in procurement
25 during the pandemic in very calm, measured and fair



1 terms. You haven't highlighted the pressure that you
2 and your team were under; I'm sure it was severe and
3 sustained. So thank you for all that you did during the
4 pandemic and thank you for the help that you've given to
5 the Inquiry.
6 I think you must be a stenographer's dream witness,
7 if I may say so.
8 **THE WITNESS:** I've tried not to go too fast.
9 **LADY HALLETT:** Thank you very much indeed.
10 **THE WITNESS:** Thank you.
11 **LADY HALLETT:** I shall return at 1.50.
12 **(12.47 pm)**
13 **(The Short Adjournment)**
14 **(1.50 pm)**
15 **LADY HALLETT:** Mr Stoate.
16 **MR STOATE:** Thank you, my Lady. Good afternoon. The next
17 witness is Matthew Style. If he could please be sworn.
18 **MR MATTHEW STYLE (affirmed)**
19 **LADY HALLETT:** I hope you were warned we wouldn't get to you
20 until this afternoon, Mr Style.
21 **THE WITNESS:** I was warned, my Lady. Thank you.
22 **Questions from COUNSEL TO THE INQUIRY**
23 **MR STOATE:** Thank you. Could you please give the court your
24 full name.
25 **A.** I can. My name is Matthew David Style.



1 **Q.** Thank you.
2 **A.** If I may, given that this is the first time I've
3 appeared before the Inquiry, just take a moment at the
4 start to express my condolences to all of those who lost
5 loved ones during the pandemic and all of those who were
6 affected and endured hardships in other ways during the
7 pandemic.
8 And just to add that during the period that we're
9 looking at today, I was a member of NHS staff at that
10 time and I was particularly conscious of the pressure
11 and the, kind of, profound sacrifice of many of my
12 colleagues at the front line during that period -- I was
13 not employed at the front line myself -- and I just
14 wanted to mark that with your permission at the start,
15 if I may.
16 **Q.** Thank you very much indeed.
17 You have very helpfully provided the Inquiry with
18 a witness statement, the reference for which is
19 INQ000513708. It's in fact your fourth statement, and
20 for the purposes of this module, the corporate statement
21 for DHSC concerning oxygen and ventilation; is that
22 right?
23 **A.** That's correct, counsel.
24 **Q.** It's 39 pages long, and is it true to the best of your
25 knowledge and belief?



1 **A.** Yes, it is.
2 **Q.** Thank you very much.
3 Just by briefly by way of background, you first
4 became a civil servant in 2001 and joined the Senior
5 Civil Service in 2008; is that correct?
6 **A.** That is correct, counsel.
7 **Q.** You have been the Director General of the NHS Policy and
8 Performance Group, now known as the Secondary Care and
9 Integration Group, since you joined DHSC in November of
10 2021; is that right?
11 **A.** That's also correct.
12 **Q.** You've just given us an insight that you were not in
13 fact at DHSC for the majority of the pandemic; you tell
14 us you were in the NHS?
15 **A.** That's correct.
16 **Q.** But you do say that the Medical Technologies
17 Directorate, about which Mr Stirling gave some evidence
18 earlier, did transfer to your responsibility in October
19 of 2022, after the closure of the Oxygen, Ventilation
20 and Devices Programme we've been discussing. Is that
21 right?
22 **A.** That's correct.
23 **Q.** So well placed, I hope, to answer some of the questions
24 that I've got for you, which in large part sit around,
25 to some degree, that of Mr Stirling's evidence. A bit

1 about preparedness beforehand and a bit about looking to
2 the future.
3 Turning then to the preparedness and pre-pandemic
4 ventilator capacity. You say in your witness statement
5 at paragraph 30 that previous modelling had been
6 undertaken that included estimating ventilator demand at
7 various periods before the pandemic to improve DHSC's
8 knowledge and understanding, including, for example, you
9 say, consideration of recommendations about bringing
10 ventilators that had been placed in storage back into
11 use; is that right?
12 **A.** That's correct.
13 **Q.** Can we look at one of those, please, one of the
14 documents that informs that. INQ000514140.
15 This is a document to which you very helpfully
16 referred us, a report of the Swine Flu Critical Care
17 Clinical Group and key learning for future surge
18 planning. If we could look at page 33 of that document,
19 please.
20 This is the section entitled "Equipment issues" and
21 looking at ventilators in particular. It says at
22 paragraph 10:
23 "Where plans include the following issues then it is
24 important that they have been tested and local actions
25 identified". Firstly:



1 "Using reserve ventilators. Where ICUs are planning
2 to use ventilators that have been kept in storage after
3 being replaced then there should be attention to
4 ensuring that mothballed equipment has been recently
5 serviced and that any specific equipment required for
6 their re-use is available in sufficient numbers to
7 enable repeated usage. As current staff may not be
8 familiar with this equipment there should be
9 arrangements made to update training, and also provision
10 of appropriate paperwork instructions."
11 Pausing there, I don't know whether you heard
12 Mr Stirling's evidence, you're probably familiar with
13 the content of it?
14 **A.** I am, I am.
15 **Q.** It touches there, doesn't it, upon some of what he was
16 talking about?
17 And then again, looking at equipment from outside
18 ICUs:
19 "If introducing equipment and devices that are not
20 normally used by existing critical care staff, there
21 should be systems created to provide appropriate
22 [updated] training sessions."
23 And gives an example of, including instructions in
24 the use of anaesthetic machine ventilators.
25 In your statement, Mr Style, you note there's -- an



1 increasing capacity in terms of these types of devices
2 was not required during the 2009-10 swine flu pandemic;
3 is that right?
4 **A.** That's correct.
5 **Q.** Ventilator baseline capacities were therefore not
6 further tested; is that right?
7 **A.** That's correct.
8 **Q.** And in fact, an increase in the baseline capacity was
9 not considered necessary in a pandemic influenza
10 scenario, is that right?
11 **A.** Yes. I should add that of course not -- I mean, this is
12 implicit in the conversation -- in my evidence at the
13 very start. I was not responsible for these issues at
14 the time so I am piecing together the history from going
15 back and looking at documents that the department holds,
16 as it were, but as you have set it out, as it is in my
17 statement. Yes, that's my understanding.
18 **Q.** You've done so, if I may say so, very helpfully. The
19 fact that it wasn't considered necessary to increase the
20 baseline capacity of ventilators, you say, was based on
21 specific learning around how many ventilators were
22 needed for events such as the 2009 swine flu pandemic.
23 Is that your understanding from looking back across
24 these documents?
25 **A.** That is my understanding, yes.



1 **Q.** One other document to which you refer us is this one,
2 please, INQ000057495. I've somewhat plonked you in the
3 middle here --
4 **A.** I'm familiar with it.
5 **Q.** I hope you would be. It's an NHS briefing entitled "NHS
6 Surge and Triaging", a briefing to the Chief Medical
7 Officer and Chief Scientific Adviser from September of
8 2017, which considered ventilator capacity.
9 If we see at the very bottom of that page, under
10 "Critical care surge", I wanted to see if you can,
11 please, help us with some of these figures:
12 "The baseline NHS capacity for critical care at that
13 time was said to be around 3,500 ventilated beds.
14 Increased demand for critical care beds above this,
15 (nominally [it says] by doubling capacity up to 7,000
16 ventilated beds) can be met through using areas within
17 hospitals that are not usually used for long-term
18 provision of such care (eg, theatre recovery areas,
19 operating theatres ...)"
20 Just pausing there, you note in your statement and
21 we've already hearing, that DHSC and NHS England, as you
22 say, had a limited understanding about the number of
23 ventilators available at that time; is that correct?
24 **A.** Yes, that's correct.
25 **Q.** Looking at this, and the other document we looked at,

1 you say that these previous studies regarding the impact
2 of an influenza pandemic hadn't considered it necessary
3 to increase, as we've discussed, and you say this:
4 "Ventilator availability was not therefore
5 identified early on as a priority given the exercises
6 that fed into pandemic planning. This was [you say]
7 also the reason why data on ventilator availability was
8 not available until the end of February 2020, as the
9 Department did not receive regular data on equipment
10 stock levels from NHSE prior to the pandemic ..."
11 Mr Style, we've heard from various witnesses now,
12 Sir Gareth Rhys Williams and Mr Stirling, we're going to
13 hear from another one this afternoon,
14 Professor Moonesinghe, about the lack of available data
15 as the pandemic hit in 2020, upon how many ventilators
16 there were, where they were, the condition they were in,
17 what model, what age. All the things you might think
18 become so crucial, in particular to clinicians.
19 Looking at this and this use of the words "nominally
20 by doubling capacity" even back then, in 2017, what has
21 been done by the department looking forward to address
22 this issue and understanding of the baseline ventilator
23 capacity, where it is, what state it's in?
24 **A.** Thank you. I think I would draw out three things in
25 respect of how the situation has changed since



1 February 2020. But before I do that, if I may, I --
2 because I -- as I say, I've been back and looked in some
3 detail at the documents that are available and are on
4 the records, and I think the overall picture, as you
5 say, is the whole report from July 2010 talks about the
6 need to be able to double critical care capacity but
7 there are no numbers specified within the whole report
8 as to what levels that would -- you know, that would
9 entail.
10 The memo to which you refer and which remains on the
11 screen from September 2017, as you say, has the figures
12 3,500 and 7,000, and there were no recommendations in
13 this memo for a baseline increase. That's the basis on
14 which I make the statement I do in my statement.
15 But to your question, the three things I think that
16 are different -- I hope I'm meeting the stenographer's
17 needs, by the way, sorry -- to your question of the
18 three things that are different today compared to
19 February 2020, the first is I believe that Ramani --
20 forgive me, sorry, Professor Moonesinghe, from whom
21 you'll hear later -- refers in her statement to an
22 annual sort of survey and census of ventilator capacity
23 that NHS England now carries out. Clearly a lesson
24 learned and implemented there.
25 The second is that in your -- in the evidence given



1 by Mr Stirling to the Inquiry earlier today, Mr Stirling
2 talked about the creation of the Medical Technologies
3 Directorate in the Department of Health, and one of the
4 activities that that directorate has undertaken, working
5 with NHS Supply Chain, is the establishment of a system
6 called NETIS. I believe that stands for the National
7 Equipment Tracking and Inventory System. But this
8 a system, an online system, that now enables us to track
9 the make, model, and location of key bits of medical
10 equipment across the NHS estate.
11 And as I say, that system is in the process of being
12 rolled out, it's already in use in some trusts but not
13 with full compliance as yet. The nature of
14 the -- (overspeaking) --
15 **Q.** Just pausing there, is that wider than just ventilators
16 and associated consumables? It's medical technology
17 more generally --
18 **A.** It is -- it is wider than just ventilators and
19 consumables, yes. I couldn't give you, as I sit here
20 now, the full -- the list of -- I mean, Mr Stirling in
21 his evidence refers to the sheer scale of the medical
22 technology supply chain and industry, but it is, you
23 know, a reasonably comprehensive system.
24 So I think, you know, that would be the second thing
25 I would mention that I think would put us in a better



1 place this time round to be able to understand the sort
2 of starting point, and indeed, to be able to tackle some
3 of the issues that you referred to with Mr Stirling
4 around ensuring that, as far as possible, where it was
5 necessary to provide additional equipment, that matched
6 the experience and expectations and familiarity levels
7 in existing organisations, because we'd know what they
8 already had, as it were.
9 So, sorry, that would be the second thing that
10 I think would lead to an improved position.
11 And then the third is to say that, as my evidence
12 makes clear, ventilators -- surplus ventilators were
13 acquired during the pandemic, which have subsequently
14 been distributed to NHS trusts, and we know,
15 I believe -- I may get the exact figure wrong, but
16 approaching some 7,000 ventilators have been distributed
17 to the NHS from the stockpile. And clearly we know
18 where those have gone and the make and model of those
19 ventilators.
20 Which I think speaks to another of the sort of
21 lessons learned to which I refer in my statement, and to
22 which indeed Mr Stirling referred earlier, which is that
23 balance between holding resilience centrally and holding
24 resilience at the front line, so that that process of
25 ensuring that clinicians can be familiar with the

1 equipment they're holding in reserve, can pay due
2 attention to its maintenance, and to the issues of
3 inoperability and so on. So, as I say, that resilience
4 has been transferred from the stockpile out into NHS
5 trusts.
6 **Q.** Thank you very much indeed.
7 Just before we turn off this document, can I ask you
8 about one other aspect of it, please. It's -- if you
9 can assist, given the context of your evidence that
10 you've told us.
11 Can you see halfway down the second paragraph that's
12 on the screen now, the sentence beginning:
13 "The trigger point within surge plans for moving to
14 the withdrawal of permanent ventilation would be when
15 there is no more capacity and increasing numbers of
16 patients who are likely to require it. This could be
17 nominally set ..."
18 Again, that use of the word "nominally" we've seen
19 twice there, just pausing there, would that be something
20 less likely to feature, now you've got a handle, as it
21 were, on all the types, numbers, makes and models of
22 ventilators across the system?
23 **A.** So I think I would say that I think we are better
24 prepared in this respect now than we were in
25 February 2020. I think beyond that, I am here as



1 a corporate witness today in respect of Module 5 and the
2 procurement activities undertaken, I'm perhaps not the
3 best witness to speak to the issues of pandemic
4 preparedness in the round. I think that best falls,
5 that best falls to others.
6 I would note that I believe that the September 2017
7 document was a draft document at that time. I have not
8 seen, in the evidence that has been available to me, any
9 further, you know, updates to the documents or variants
10 prepared since.
11 **Q.** All right. We'll stop there in relation to that topic
12 then and I just want to turn briefly with you, if I may,
13 to complete our understanding of the certain aspects of
14 the procurement of ventilators. Briefly, please, on the
15 numbers which are procured. This is your paragraph 51.
16 We know that in March there were the tentative
17 estimates of 6,000 to 8,000. You tell us that by
18 9 April there were 9,600 ventilators available, yes.
19 **A.** Yes.
20 **Q.** By 14 April, the department had procured a total of
21 10,993 mechanical and 19,338 non-invasive ventilators,
22 although many hadn't arrived in the UK. We're talking
23 significant numbers at significant speed here, aren't
24 we?
25 **A.** Yes, and I think -- and again, I think this came up in



1 your evidence with Mr Stirling earlier, and it's perhaps
2 easier for me to say it, as someone who wasn't involved
3 in the programme at the time, I think once the programme
4 was mobilised on 3 March, you know, the programme was
5 very successful in mobilising quickly to expand supply
6 by very significant volumes --
7 **Q.** Just to --
8 **A.** -- and that people involved at the time can be duly
9 proud of their efforts in doing so.
10 **Q.** Just to complete the picture, INQ000494406, a slide that
11 you provided within your evidence about these numbers.
12 So we've gone through April. Now we're at 19 May.
13 This is a stocktake of ventilators, circulated on that
14 date, the 19th.
15 By now we can see that the total number of
16 ventilators, including mechanical, NIV, and CPAPs is in
17 the region of 34,800 comprising, as we see, 13,000
18 mechanical, a bit more than 11,000 bilateral or BiPAP
19 ventilators, and 10,100 CPAP ventilators, compared to
20 a pre-pandemic stock of 4,954 ventilators.
21 Can I ask you this question briefly: we can see the
22 numbers that were achieved and we can see the speed in
23 which they were achieved. Does the department agree
24 with -- do you, Mr Style, agree with Mr Stirling and
25 Professor Moonesinghe about the variably quality of



1 which those ventilators were, given the speed?
2 **A.** I think that's, it's clear from the evidence that there
3 was indeed variable quality. That was clear from
4 Mr Stirling's statement earlier and, indeed, in the
5 documents I reviewed and have exhibited against my
6 statement there are -- there was evidence of that very
7 much from frontline colleagues at the time.
8 **Q.** Thank you.
9 One aspect of this procurement was that of
10 ventilators from overseas. This is your paragraph 85.
11 I want to show you one document, I believe. It was
12 referred to in Mr Wald KC's opening. INQ000493942.
13 This is a document entitled "Ventilator Sifting Wash-up
14 Report" by the Department for International Trade. It's
15 dated 24 April 2020. And within here, there is
16 a section reflecting upon the period at the end of March
17 and the beginning of April and what's said is this:
18 "... there was huge international demand, greatly
19 outstripping supply. Speculators, opportunistic
20 intermediaries and individuals had piled in, trading up
21 prices exponentially. Some of the units that we looked
22 at had changed ownership over five times in the past two
23 weeks. The prices quoted were on average triple the
24 usual retail price and at the peak of the market, many
25 times over."

1 And then this phrase which I think is quite vivid:
2 "We had entered a ventilator procurement 'Wild
3 West' ..."
4 Now, appreciating you -- that wasn't part of your
5 remit at the time, is that a -- looking now at the
6 procurement of ventilators from overseas, is that
7 a picture that you recognise, having looked back across
8 the documents?
9 **A.** I think it's undoubtedly the case. And again, I think
10 that was clear from Mr Stirling's evidence and from the
11 exhibits attached to my statement that it was an
12 incredibly challenging commercial environment, both
13 domestically and internationally, and colleagues, both
14 in the Department of Health and Social Care, from the
15 Government commercial function more broadly, and indeed
16 from the then Foreign and Commonwealth Office worked
17 together and went to, kind of, great lengths in order to
18 secure supply and sort of protect the taxpayer in what
19 was a very, as I say, a very challenging commercial
20 environment.
21 I wouldn't comment on the language used by the
22 author of this particular report.
23 **Q.** Fair enough. But you say at your paragraph 85, just in
24 terms of how this process happened:
25 "... officials from the Department indicated that



1 they would be asking [the Treasury] for delegated
2 authority ... to order ventilators ..."
3 And that included on condition that there was an
4 assessment of international suppliers in writing from
5 both the FCO, as you've said, and DHSC, to show that
6 they were satisfied with the level of any risk of that
7 purchase. Is that right?
8 **A.** That's right. So teams worked together very closely,
9 both to, you know, receive offers and sort of
10 expressions of supply, and make sure those were acted on
11 swiftly, and UK-based teams worked with FCO colleagues
12 both to obtain documentation as to the sort of clinical
13 specification and for the compliance with standards of
14 any offers to try and extract information as to the
15 location of devices, where they were being held and so
16 on. And indeed, as I think the exhibit you just put on
17 screen makes clear, often you were dealing with
18 re-sellers rather than manufacturers, indeed, possibly
19 multiple re-sellers.
20 And again, FCO colleagues were instrumental in
21 trying to get to the bottom of that thread, as it were,
22 and helping to inform judgements being made about the
23 credibility of those offers.
24 And again, as someone who wasn't involved in the
25 programme at the time, in going back and reviewing the



1 evidence -- and again, you can see this in my
2 exhibits -- what is striking is the speed with which
3 colleagues were having to make judgements and liaise
4 between, for example, Mr Stirling, who appeared before
5 the Inquiry earlier today, his accounting officer and
6 finance colleagues in the Department, Treasury
7 colleagues, FCO colleagues both in London and in-country
8 to sort of pool information and make those judgements
9 very quickly, as prices were escalating mid-deal. Some
10 of these examples I referred to in my statement, as you
11 know.
12 **Q.** Can we look briefly at one of those examples, please.
13 Your paragraph 91. Just talk us through this if you
14 would, please. This was at the end of March 2020. You
15 were made aware of an offer from a company in Malta, the
16 number of ventilators doesn't matter, but just talk us
17 through this example, if you would, and then I'll ask
18 you some questions about what it might tell us.
19 **A.** Sure. So on 26 March 2020, the FCO made the department
20 aware of an offer from a company in Malta, I was about
21 to say its name but I probably ought not to, to supply
22 ventilators initially for fewer than is stated in my
23 statement here, but the number available rose to 3,000,
24 you know, very -- the number purported to be available
25 rose to 3,000 in the first, you know, stages of those



1 discussions.
2 There was extensive contact with the company in
3 Malta from the department, but also Stuart Gill, who at
4 the time was the British High Commissioner in Malta had
5 contact both with the company and with the Foreign
6 Minister in Malta in order to do due diligence and
7 arrive at a view as to the reputability of the company,
8 which appeared to sort of stack up in the first
9 instance.
10 It then became clear, as the deal evolved, certain
11 bits of information came to light, for example, that the
12 stock was held in China and would be transferred to the
13 UK from Shanghai using Qatar Airways. And the original
14 agreement was that a deposit would be paid of just short
15 of €40 million.
16 Once that had been paid, there was then -- two
17 changes were then made to the deal on the table. One
18 was to seek an increase in prices. I believe that was
19 on 1 April that that approach was made, and to require
20 an immediate letter of credit.
21 We -- the department proceeded at the time working
22 with the Government legal department and its bankers,
23 NatWest, to prepare a letter of credit, but made clear
24 that the usual requirements in terms of confirmation of
25 a delivery schedule and the location of the items for

1 inspection and so on would be needed.
2 Those were not forthcoming, and on both 4 April,
3 I believe, and 10 April, the department repeated its
4 requirements to the supplier. Ultimately, given that
5 there was no confirmation of delivery date and no
6 confirmation of the other details requested, on 16 April
7 we informed the company that we wouldn't proceed with
8 the deal and requested a refund of the deposit.
9 That was -- a refund was ultimately obtained. As my
10 statement makes clear at paragraph 93, we did sustain
11 a loss on that deal as a result of exchange rate
12 fluctuations. I believe the loss amounted to
13 approximately 500 -- it might be £540,000. But it was
14 of that order.
15 **Q.** Yes. So if you don't mind me saying, really quite
16 a significant loss. That's not a small amount of public
17 money, is it?
18 **A.** It is not a small amount of public money. I think this
19 is a good example of the very difficult judgements that
20 the team were having make in order to secure the best
21 possible supply.
22 **Q.** So "Wild West", not your phrase, but a very difficult
23 purchasing and procurement environment from abroad.
24 You've very helpfully talked us through one example and
25 the loss that was sustained.



1 Do you, looking now, and does the department, think
2 that in advance of any future pandemic, it's the
3 stockpile question, that there should be a stockpile of
4 equipment like ventilators and associated hardware.
5 Appreciating there's a balance there, would that help to
6 lessen the risk of situations such as this one arising?
7 **A.** So the first thing I'd say is I believe we are in a more
8 resilient position today than we were in February 2020,
9 not least because there was increased resilience held,
10 you know, at the -- locally, as it were, as opposed to
11 in a national stockpile. And clearly, if we were
12 having -- if we didn't have such a significant need to
13 procure additional ventilators in the sort of commercial
14 environment that we've rehearsed, then we would be, sort
15 of, less exposed to potential, sort of, losses and
16 exploitation in that environment.
17 But there is, of course, a balance to be struck, and
18 I think Mr Stirling in his evidence this morning
19 referred to the -- you know, the sort of cost of
20 storage, if you hold a national stockpile. So, in
21 a sense, it's not a sort of -- you know, this is not
22 a one-way balance as it were.
23 And also, the risks of -- you know, the difficulty
24 of maintaining equipment, ensuring that the equipment,
25 you know, keeps pace with -- ensuring the equipment



1 keeps pace with changes in clinical preference and
2 indeed technological change as most effective equipment
3 to hold.
4 And indeed, and again this is -- you'll have heard,
5 the Inquiry, and perhaps even you, counsel, will have
6 heard other evidence from colleagues in the department
7 about, you know, you don't know specifically which
8 equipment you're going to need depending on which
9 pandemic is going to hit you. And it's particularly
10 striking in this case, I think, to -- and again, this
11 was rehearsed extensively in the technical report which
12 is an exhibit to my evidence, and indeed Mr Stirling
13 referred to it earlier today -- the clinical model of
14 care evolved significantly throughout the pandemic. At
15 the start of the pandemic, the focus was very much --
16 and indeed the procurement focus of the programme -- was
17 very much on mechanical ventilation.
18 Towards the end of the -- or even after the very
19 first phases of the pandemic, the emphasis, clinical
20 practice -- and again, technical report rehearses
21 this -- clinical practice switched towards non-invasive
22 ventilation.
23 So we could have had a sort of, you know, huge, huge
24 stockpile of mechanical ventilators and we may not have
25 had, you know, as many CPAP machines as we eventually



1 decided that we needed. So I think it is very difficult
2 to foresee every single requirement for every single
3 sort of potential pandemic, and hold those centrally in
4 a kind of physical stockpile.
5 These are very difficult judgements. I think, you
6 know, one of the lessons of the -- one of the lessons of
7 the pandemic is that those judgements need to be
8 subjected to, you know, very careful scrutiny and to be
9 made and subject to very regular review, as it were.
10 And I think that's sort of critically important because
11 there is no -- there was no easy answer to this
12 question.
13 **Q.** Thank you very much.
14 Turning briefly, please, to one other aspect of your
15 statement and that's about attempts to mitigate
16 fraudulent behaviour in the procurement of this
17 equipment.
18 One of the documents to which your statement refers
19 is INQ000514235.
20 Just going to talk through it very briefly, please.
21 This is a letter from David Williams, second permanent
22 secretary to the DHSC, to the Public Accounts Committee
23 in October of 2020, and a response to requests to write
24 to that committee setting out instances of fraud when
25 attempting to procure ventilators. Just a couple of

1 points that are made.
2 Can you see here in this last part on the screen in
3 the last section:
4 "Often it was difficult to differentiate between
5 over-optimism in complex supply chains and deliberate
6 fraud. The due diligence process undertaken to test the
7 proposed was designed to protect both against
8 over-optimism and fraud."
9 This is effectively suppliers claiming to have stock
10 that they didn't; is that right? That's what's being
11 aimed at here?
12 **A.** I believe so, yes. And this is the -- the PAC -- the
13 letter to the PAC from David Williams, then I think
14 second permanent secretary at the time, to the PAC, yes.
15 Yes, that's my understanding.
16 **Q.** Next section, on "Fraudulent Documentation", we can see
17 that:
18 "In order to assess the validity of offers, the
19 Department requested documentation from potential
20 suppliers, [including] CE certificates ..."
21 And other things.
22 **A.** Yes.
23 **Q.** You say:
24 "Whilst some of these documents could be
25 independently identified ... others were more



1 challenging ..."
2 Then this:
3 "There were a very small number of cases where
4 clearly fraudulent documents were supplied: [and] these
5 were not taken forward."
6 Mr Style, for obvious reasons, I don't want to get
7 into the specifics of any particular case, but I wanted
8 to get the department's view on, overall, to what extent
9 was fraud something that impacted upon the procurement
10 effort of this type of medical equipment?
11 **A.** Well, look, we -- clearly, fraud on any scale is
12 something of very significant concern, and I take my
13 responsibilities -- as would David Williams, as the
14 accounting officer responsible at the time, have taken
15 our responsibilities -- for safeguarding taxpayers'
16 investment very seriously indeed.
17 Overall, I think this particular procurement
18 programme was less vulnerable to fraud than some others
19 may have been in these circumstances, simply because it
20 was, sort of, easier to detect, possibly, some, sort of,
21 clearly fraudulent or clearly over-optimistic offers, as
22 it were. If people were suggesting that they could
23 suddenly supply three times the typical annual output of
24 a manufacturing facility of a ventilator, when it was --
25 clearly didn't withstand scrutiny, as it were.



1 As others have rehearsed in the evidence, these
2 are -- ventilators are complex pieces of, sort of, you
3 know, clinical engineering, and you've got to have
4 a certain degree of, kind of, established capability in
5 order to be a credible supplier. And I think that helps
6 to limit the programme's exposure to fraud overall. And
7 as the letter to the PAC says, there was only one
8 sophisticated suspected case of fraud identified.
9 But as I say, I think that's -- in some ways, that's
10 a testament to the efforts of staff in London,
11 in-country, and so on, to perform as much due diligence
12 as they sort of possibly could. I won't go over the
13 details of that, I rehearsed it a little bit earlier on.
14 And I think it was those teams working together that
15 helped to, you know, minimise these risks of fraud very
16 successfully during the programme.
17 **Q.** Thank you.
18 Just one question now, please, and I'm sorry to have
19 moved you round sort of in short time between a whole
20 series of topics, but you've been very helpful in your
21 answers, if I may say. It's a topic upon which your
22 witness statement and the evidence of Mr Stirling this
23 morning touches. It's that of clinician choice, and
24 it's something you touch upon in your paragraph 155.
25 **A.** Mm.



1 **Q.** First, there's this acknowledgement which I'm just going
2 to read to you:
3 "Given clinicians can have strong preferences for
4 particular equipment models and brands, greater account
5 could perhaps have been taken of these preferences in
6 determining the appropriate balance between stimulating
7 the production of existing designs and developing novel
8 designs."
9 So you explicitly acknowledge that.
10 But looking forward, talking about
11 interchangeability and interoperability, what's the
12 up-to-date position as far as the department is
13 concerned?
14 **A.** Thank you. I think the first thing I'd say is I think
15 my colleagues working on these issues at the time were
16 right to pursue all of the available options. The scale
17 of the demand projections -- and again, I believe,
18 Mr Stirling rehearsed you with this earlier -- the scale
19 of demand projections was such that I'd think colleagues
20 in the programme and their colleagues in the Cabinet
21 Office working on the parallel programme were right to
22 pursue all options. And, you know, that was --
23 that's -- that's, as I say, that was the right thing to
24 do.
25 I do think it is very important to bear in mind the

1 issues of familiarity with machines' interoperability
2 and so on. I think the key thing I would point to there
3 is the sort of inventory and tracking system I referred
4 to earlier. So we would be starting -- were we to find
5 ourselves in this position again, we'd be starting with
6 a better understanding of which devices, of which make
7 and model were in use, in which organisations, as it
8 were, and therefore we would be better able to ensure
9 that we were sort of matching clinical preference and so
10 on.
11 But I think the other thing I would say is the
12 importance of these issues, the importance of these
13 issues of interoperability and familiarity with a range
14 of devices, you know, the pandemic experiences
15 absolutely shone a light on that, and I think those
16 clinical teams at the front line in thinking about their
17 own resilience will clearly be much more mindful of
18 those issues, you know, based on those lessons than they
19 kind of may have been previously.
20 **MR STOATE:** Thank you very much.
21 My Lady, those are my questions. There are some
22 more --
23 **LADY HALLETT:** Thank you very much.
24 Ms Banton, I think you've got some questions.
25 Over that way.



1 **THE WITNESS:** Thank you.
2 **Questions from MS BANTON**
3 **MS BANTON:** Thank you, my Lady.
4 Just for some context, Mr Style, I represent FEMHO,
5 and in your capacity over seeing critical aspects of
6 pandemic response, particularly related to ventilators
7 and oxygen therapy devices, it's crucial to explore how
8 considerations of equality impacted decision-making
9 processes. FEMHO would like to understand the extent to
10 which the needs of black, Asian and minority ethnic
11 healthcare workers were considered in procurement and
12 policy formulation, helping to shape recommendations for
13 improving resilience and equity in future health crises.
14 So my first question is in relation to paragraphs 40
15 and 61 of your witness statement, where you describe the
16 processes and shifts in strategy regarding the
17 allocation and management of ventilators and oxygen
18 therapy devices.
19 How did the national ventilator allocation panel
20 ensure compliance with the Public Sector Equality Duty
21 when making allocation decisions?
22 **A.** Thank you. And it's obviously a very important issue,
23 and one of those on which the pandemic has, you know,
24 shone a particular light on the need to consider the
25 differential impact that, you know, different health



1 emergencies will have on populations.
2 I think two things are important in response to your
3 question. The first is that the decisions that the
4 programme, through the national ventilator allocation
5 panel were making were first and foremost informed by
6 the clinically-led projection. So, as you will know,
7 you know, as you will understand much better than me,
8 that understanding about the differential impacts of the
9 virus on different sections of the population, including
10 the particular impacts on people from ethnic minorities
11 as well as other factors such as age and, sort of,
12 geographical factors and so on, that understanding
13 evolved over the pandemic, but as that evolved, that
14 increasing understanding was then being fed into the
15 clinical work that was being done to make projections of
16 future demands, and those projections of future demand
17 were then fed into the allocation panel in order to make
18 its decisions.
19 Similarly, and again as Mr Stirling said earlier
20 today, the key inputs into that allocation process came
21 from regional medical directors and their teams, who --
22 I mean, I was not part of the process at the time, but
23 I have worked extensively with regional medical
24 directors over the years, and I think that they, you
25 know, were sort of well placed to have that sort of very



1 up-to-date connection with differential pressures and
2 the very localised impacts on particular communities and
3 therefore on particular, you know, hospital trusts and
4 healthcare providers that wouldn't necessarily have been
5 picked up in sort of aggregate data, but the national --
6 the relevant medical director would have known that
7 particular pressures were being experienced in that
8 institution because of the demographics of its sort of
9 local population.
10 So I think those are the ways in which the factors
11 you describe, and in particular the differential impact
12 on people from ethnic minorities would have been
13 appropriately reflected in and, sort of, had input into
14 decisions made by the national ventilator allocation
15 panel.
16 **Q.** Right. Okay. Thank you.
17 I'll ask my next question. In the governance of the
18 O2VMD&CC programme that you mentioned, various
19 decision-making bodies and their roles, and this is at
20 paragraph 60 and then 63 to 66, so in respect of that
21 I've just got just two points to raise.
22 The first one, how were representatives from ethnic
23 minority-led healthcare providers included in these
24 forums, if at all, to ensure their specific needs were
25 addressed?

1 **A.** From the top of my head I couldn't tell you whether or
2 how specific representations from ethnic minority
3 communities were reflected in the governance of the
4 programme. I think the thing which I -- if I may, one
5 of the things I would draw attention to -- and
6 I referred earlier to the fact that this is something in
7 which learning and understanding has advanced
8 significantly as a result of the pandemic experience,
9 you know, we have since established, as you will no
10 doubt know, the NHS Race and Health Observatory under
11 the leadership of my colleague Professor Habib Naqvi,
12 and that's now a very well established channel for
13 ensuring that policymakers, operational decision makers,
14 can access analysis insight and representations to make
15 sure that's absolutely at the heart of decisions that
16 are now -- as they are now made across the system.
17 **Q.** Right. So was there any mechanism to gather input from
18 healthcare providers serving predominantly ethnic
19 minority communities to inform procurement strategies?
20 **A.** I think the principal vehicle for doing so would have
21 been through the -- both the clinical and operational
22 representatives of -- from NHS England. So for example,
23 Dr Dame Emily Lawson, as SRO for the programme, and
24 Professor Sir Keith Willett as the National Director for
25 Emergency Preparedness in NHS England were both



1 members -- were both represented in programme
2 governance, and, and both Keith and Emily would have
3 been in daily, several times daily contact with
4 frontline providers and therefore had that very sort of
5 swift intelligence as to the differential impacts and
6 particular providers struggling to meet the needs of
7 particular communities. It would have been through
8 those mechanisms that that intelligence was available to
9 the programme.
10 **Q.** Right. My next question. Considering the preparation
11 for future health threats and lessons learned going
12 forward, and not just only from the pandemic, as
13 discussed in paragraphs 166 to 168 of your witness
14 statement, the first point is, how does the government
15 assess its effectiveness in protecting ethnic minority
16 healthcare workers during the pandemic?
17 **A.** Well, as I said, if I may, as I said at the start of my
18 evidence, I was a member of NHS staff at the time and
19 I was particularly conscious of the challenges faced by
20 colleagues at the front line, and, as you say, some
21 people were more affected than others, and experienced
22 greater challenges. I think -- it's not something on
23 which I am -- I mean, I'm not the appropriate witness to
24 speak to this in detail, but what I would say, my
25 understanding, is that colleagues in the various



1 different sort of operational cells responding to the
2 pandemic acted as swiftly as they could once the -- once
3 the sort of data and, you know, intelligence began to
4 show that there were -- that some staff were
5 experiencing, you know, greater challenges than others.
6 And I saw, again, colleagues in NHS England at the time
7 acting, sort of, swiftly to, sort of, very rapidly learn
8 those lessons and change the way they were, you know,
9 leading and managing staff, including to take account of
10 the preference -- the preferences of individual
11 employees about their own deployment and so on through
12 the pandemic, precisely to ensure that, you know, a full
13 account could be taken of the sorts of differential
14 impacts to which you refer.
15 **Q.** And following on from that, what sort of specific
16 measures are being implemented to ensure better
17 preparedness and protection for these workers in
18 response to future health crises?
19 **A.** Forgive me. I mean, this is a critically important
20 issue. It's not something that I am the best person to
21 speak to, as a corporate witness for the procurement
22 module, as it were. But I -- it's clearly a critically
23 important issue.
24 **Q.** Right.
25 How, would you say, are the lessons learned being



1 integrated into current and future policies to enhance
2 the resilience and safety of the healthcare system,
3 particularly for ethnic minority communities?
4 **A.** Well, I think I would go back to my overall assessment
5 is that we have -- we now have in place a system that
6 allows us to much better understand the resilience of
7 all of our hospital trusts across the country. So we
8 would -- were we to find ourselves in this -- in
9 a similar position again, we would start from -- with
10 a much better understanding of resilience.
11 We also, as I said, not least through the work of
12 the Race and Healthcare Observatory, you know, we have
13 established mechanisms to ensure that we're thinking
14 actively about differential impacts on different
15 communities, including different communities within our
16 staff. We're much more conscious of it, we have
17 established mechanisms to ensure that's reflected in our
18 policy making and our operational decision making.
19 **MS BANTON:** Thank you.
20 Thank you, my Lady. Those are my questions.
21 **LADY HALLETT:** Thank you, Ms Banton, very grateful.
22 Mr Stoate, I think that completes the questions for
23 Mr Style.
24 **MR STOATE:** It does.
25 **LADY HALLETT:** Thank you very much indeed, Mr Style. I'm

1 very grateful to you.
2 **THE WITNESS:** Thank you.
3 **LADY HALLETT:** Mr Stoate, I'm in your hands, it's early for
4 a break but we're making very good progress.
5 **MR STOATE:** Yes.
6 **LADY HALLETT:** It's entirely up to -- you're calling the
7 next witness I think?
8 **MR STOATE:** I am.
9 **LADY HALLETT:** It's entirely up to you whether we get on to
10 the next one or take the break now.
11 **MR STOATE:** I can certainly make a start, my Lady, if that
12 would ...
13 **LADY HALLETT:** Mm-hm.
14 **MR STOATE:** Thank you.
15 The next witness, please, is Professor Ramani
16 Moonesinghe.
17 **PROFESSOR RAMANI MOONESINGHE (affirmed)**
18 **LADY HALLETT:** Professor, I don't think we have kept you
19 waiting too long. I hope we haven't.
20 **THE WITNESS:** Thank you.
21 **Questions from COUNSEL TO THE INQUIRY**
22 **MR STOATE:** Thank you.
23 Please could you give the court your full name.
24 **A.** Ramani Moonesinghe.
25 **Q.** Thank you. You've very helpfully given the Inquiry



1 a statement, the reference for which is INQ000518349.
2 It's 54 pages long and dated 28 October of last year.
3 Is that statement true to the best of your knowledge and
4 belief?
5 **A.** Yes, it is.
6 **Q.** Thank you.
7 Professor, the Inquiry has had some impressive CVs,
8 but yours, if you don't mind, will take a little while
9 to go through.
10 You are a doctor who has worked in the NHS for some
11 27 years, including 20 working clinically in critical
12 care; is that right?
13 **A.** Correct. Thank you.
14 **Q.** You were appointed a consultant in critical care,
15 anaesthesia and perioperative care in UCLH in 2009.
16 **A.** Yes.
17 **Q.** Where you worked in a multi-disciplinary team providing
18 care for critically ill patients in one of the largest
19 services in the UK.
20 **A.** Yes.
21 **Q.** You're a fellow of the Royal College of Physicians, the
22 Royal College of Anaesthetists and the Faculty of
23 Intensive Care Medicine, and a senior fellow of the
24 Faculty of Medical Leadership and Management?
25 **A.** I am.



1 **Q.** You were appointed as a national speciality adviser for
2 perioperative care at NHS England in 2019?
3 **A.** Yes.
4 **Q.** You started, you say, providing input on clinical and
5 operational matters related to the Covid-19 response
6 within NHS England on 2 March 2020?
7 **A.** Correct.
8 **Q.** And you were formerly appointed to the role of National
9 Clinical Director for Critical and Perioperative Care at
10 NHS England on 19 March 2020 to provide national
11 clinical leadership for adult critical care in the
12 pandemic, and in fact, still now?
13 **A.** That is correct.
14 **Q.** And are there any other positions you currently hold?
15 **A.** I am currently also the interim National Director for
16 Patient Safety for the NHS in England.
17 **Q.** Thank you very much indeed.
18 As the first clinician today talking about some very
19 complex pieces of kit, you may or may not wish to set us
20 right, but when we're talking about the procurement of
21 ventilators, and associated kit, we're talking about
22 a fairly complex range of specialist equipment, aren't
23 we, for treating critically ill patients; is that right?
24 **A.** That is correct.
25 **Q.** Could you give us a brief but, sort of, overall flavour



1 of what is that stuff that's needed?
2 **A.** Thank you. So critical care treats patients who either
3 have organ failure or are at risk of it. The organ
4 failure that we most commonly associate with critical
5 care is respiratory or lung failure and the treatment
6 for respiratory failure is either mechanical
7 ventilation, where one places a breathing tube into
8 a sedated patient and then the mechanical ventilator
9 either fully breathes for the patient or supports the
10 patient's own breathing effort.
11 The next step down from that is non-invasive
12 ventilation, where a patient would use a tight-fitting
13 facemask, and air and oxygen would be blown into them
14 through that.
15 The next stage down from that is continuous positive
16 airway pressure, or CPAP. Some people may be familiar
17 with CPAP because we use it as a treatment for
18 obstructive sleep apnoea, so many people have CPAP
19 machines at home, but in the hospital environment, and
20 indeed during Covid, we also use it to treat respiratory
21 failure, again through a tight-fitting facemask, or
22 sometimes a helmet or hood.
23 Critical care also treats organ failure or risk of
24 organ failure of all our other organs. So that includes
25 our cardiovascular system, usually with drugs but

1 sometimes with mechanical devices; the kidneys, with
2 renal replacement therapy, a type of dialysis, and so on
3 and so forth.
4 And obviously critical to all of that is the staff,
5 because this is the absolute sharpest end, I suppose, of
6 what we offer in terms of ward-level care in the
7 hospital, and so the staff that work within critical
8 care have been trained to use all of these machines,
9 understand the context of failing or failed organs, and
10 provide that holistic care to the patient, and also to
11 their loved ones.
12 **Q.** Thank you. And as well as the staff, and the machines
13 themselves, there is a wide range of other consumable
14 items that go alongside that, aren't there?
15 **A.** That is correct. So capital equipment like monitors, so
16 we can see the blood pressure, and so on, continuously.
17 Consumable items. So I've already mentioned breathing
18 tubes, but the drips in our hands and all sorts of other
19 things, catheters, and so on.
20 So one of the things that we had to do at the
21 beginning of the pandemic was actually to itemise
22 everything that we might use in a critically ill
23 patient, how much we would use, how many we would need
24 per individual patient based on projections of how long
25 we thought patients would be in hospital requiring that



1 kind of help, and so on. It's a very, very complex
2 task.
3 **Q.** One of the tasks that we'll go on to discuss that you
4 had was clinical leadership in the procurement of this
5 stuff, wasn't it, that we've been discussing?
6 As part of the preparation for this Inquiry, you
7 were provided with a statement by Professor Philip
8 Banfield, chair of the British Medical Association's UK
9 Council.
10 **A.** Yes.
11 **Q.** Perhaps we can bring it up. It's INQ000562457 at
12 page 85, at paragraph 274. I've got it on the screen
13 for you there, I hope, Professor?
14 **A.** Yes.
15 **Q.** The observation of the BMA here is this:
16 "... the UK started out on the back foot in relation
17 to the supply of intensive care [ventilators], having
18 failed to sufficiently consider ventilation capacity as
19 part of pre-pandemic planning and preparedness. This is
20 because recommendations set out an Exercise Cygnus were
21 largely overlooked, despite concerns around the ICU
22 ventilator capacity then ... The failure to action the
23 recommendations meant that, at the pandemic's outset the
24 UK was already at serious risk of running out of
25 ventilators, with all 4 governments noting a very



1 significant shortage of ventilators compared to
2 estimated potential need. Indeed, the UK entered the
3 pandemic without enough ventilators and ICU beds for
4 normal levels of emergency and elective care; the lack
5 of a staffed ICU bed was (and remains) a frustratingly
6 common reason for the cancellation of major surgical
7 procedures."
8 Speaking from your position, do you agree with
9 all/part of that picture presented here?
10 **A.** So I can't comment on the recommendations from Exercise
11 Cygnus because I have not seen that.
12 **Q.** Okay.
13 **A.** However, there's no doubt that the NHS in England and
14 I think across the UK has lower critical care capacity
15 on a day-to-day basis than comparable countries, so OECD
16 countries. And this has been noted previously, my Lady,
17 in Module 3 by Professor Chris Whitty and also by the
18 expert witnesses for intensive care.
19 So we went into the pandemic with approximately
20 six beds per 100,000 population in England, and today,
21 having increased our intensive care capacity as a result
22 of Covid and since then some investment being made,
23 we're at about seven.
24 But many other countries are at nearer ten or
25 eleven. And we can perhaps talk about that more later.



1 **Q.** Yes.
2 You came in -- you started providing, as you already
3 told us, clinical input on operational matters on
4 2 March, took up your formal role as national clinical
5 care director on the 19th. I wanted to ask you about
6 the picture it presented to you.
7 There was, as we've heard from others, really quite
8 a stark lack of data about the availability of all the
9 kit that we've just discussed. Do you share that
10 assessment?
11 **A.** So a stockpile was undertaken towards the end of
12 February which gave us an understanding of the number of
13 critical care beds and ventilators in the system, and
14 also the number of anaesthetic machines, which were
15 a secondary source of ventilatory equipment, not ideal,
16 but a reasonable next choice, and also gave us an
17 understanding of the number of transport ventilators,
18 which were, again, not an ideal choice for ventilating
19 critically ill patients but a means of doing so.
20 So that was organised by our emergency preparedness
21 response and -- EPRR team at the end of February, and
22 that gave us the numbers, which were, from memory, in
23 the order of around 7,000 ventilators -- I may have got
24 that slightly wrong -- of which about half were true
25 critical care ventilators and the rest were second

1 choice equipment.
2 Q. The witness statement to the Inquiry of
3 Dame Emily Lawson, chief commercial officer of
4 NHS England during the pandemic, as I'm quite sure you
5 know, says as follows -- I don't think I need to bring
6 it up -- can I just put this point to you? It's her
7 paragraph 99. She says, of exactly this time:
8 "One of the initial problems I encountered after
9 joining the team [this Mr Stirling's team] was that
10 there was no recent indication of the number of existing
11 ventilators, their make or their capabilities, available
12 to the NHS. This information was needed to inform
13 purchasing."
14 Then this:
15 "A ventilator survey had been conducted in 2017 and
16 needed updating. I had no concerns about this;
17 ventilator numbers are not data that I would have
18 expected us to routinely collect and hold nationally."
19 Did you have any concerns about the fact that there
20 hadn't been a survey since 2017, the lack of data? Was
21 it concerning to you?
22 A. I suppose by the time I got involved we already had the
23 numbers. So I think it's surprising, perhaps, that we
24 did not have more up-to-date information than 2017. It
25 is something that we have learnt from, and the NHS and



1 NHS England conducts a survey -- has conducted a survey
2 three times, so annually, since Covid, to understand the
3 number of intensive care beds, the number of ventilators
4 in intensive care units, and also our immediate surge
5 capacity as well. So beds that could be stood up
6 within 24 to 48 hours to support critically ill patients
7 should it be required outside a normal context.
8 Q. In any event, you say it was clear -- and I think we've
9 heard this now from number of people -- that there were
10 insufficient numbers of ventilators to meet the demand?
11 A. That is correct. So when I was first involved, in early
12 March, one of the first jobs that I did was to support
13 looking at lists of ventilators that we could
14 potentially procure, and I'm sure we'll talk about that
15 in a moment, and our strategy, collectively, was to map
16 every single thing that would be required to treat
17 a critically ill patient, and make sure that not one
18 individual thing would be the rate limiting step in
19 being able to treat as many people as possible.
20 So the first -- the reasonable worst case scenario
21 data that I was presented with on around 15 March 2020
22 indicated that we would need at least 30,000 intensive
23 care ventilators, and possibly many tens of thousands
24 more than that. So starting with a baseline of 7,000,
25 it was clear that we had a long way to go.



1 Q. Yes. We'll come, probably after the break, to the
2 Ventilator Challenge. We've heard a bit about that.
3 That was designed, wasn't it, to fill that gap
4 between --
5 A. Yes.
6 Q. -- supply and demand. But just before we get there,
7 I want to ask about this observation of yours that in
8 parallel with the idea of manufacturing new ventilators
9 through the challenge, you say this:
10 "... an increasing number of innovation proposals
11 were being received by NHS England, DHSC and
12 professional bodies, necessitating the [deployment] of
13 a systemic approach to evaluating", other options,
14 effectively.
15 What were those other options that were being
16 considered?
17 A. So it speaks to the ingenuity of my colleagues and the
18 effort that everyone was trying to put in to meet the
19 challenge that some very unusual ideas came to the fore.
20 For example, repurposing veterinary ventilators and
21 anaesthetic machines for humans. That was reviewed by
22 somebody from the Royal Veterinary College and
23 dismissed. The idea of splitting a ventilator, which
24 would always be used to treat a single person between
25 two people. Again, something that we considered and



1 dismissed.
2 Then the third idea was to try to make ventilators.
3 And that -- and we were -- multiple people were
4 thinking, as I say, in the same way. That is what
5 lengthily let on to the Ventilator Challenge.
6 Q. As well as speaking to ingenuity, does it also speak, in
7 your view, to a lack of preparedness?
8 A. A hundred per cent.
9 MR STOATE: My Lady, that might be a convenient moment.
10 LADY HALLETT: Certainly. I shall return at quarter past.
11 I hope you were warned about our break.
12 THE WITNESS: Yes.
13 LADY HALLETT: It depends on the witness. We do have
14 perfect witnesses occasionally, as we did this morning.
15 I shall return at 3.15.
16 (2.58 pm)
17 (A short break)
18 (3.15 pm)
19 LADY HALLETT: Mr Stoate.
20 MR STOATE: Thank you, my Lady.
21 Professor, before the break, we were just turning to
22 the Ventilator Challenge. We have already heard in the
23 Inquiry quite a lot of evidence from Sir Gareth Rhys
24 Williams --
25 A. Yes.

1 **Q.** -- with whom you worked very closely in that endeavour?
2 **A.** Indeed.
3 **Q.** So I only want to ask you about a couple of specific
4 aspects of it. One of the initial things you did as
5 part of the Ventilator Challenge was to produce a draft
6 of what you call the Rapidly Manufactured Ventilator
7 Specification. What was that?
8 **A.** It was the minimum requirement for a ventilator to be
9 made through the Ventilator Challenge. It didn't set
10 a maximum requirement; but it was the minimum clinically
11 acceptable requirement for a ventilator.
12 **Q.** And what key features were you looking for in that
13 requirement?
14 **A.** So if may, I just want to provide a little bit of
15 additional context. We've talked about how complex
16 ventilators and that type of equipment are. And
17 although as a clinician, I know quite a lot about how
18 a ventilator works, I don't know a great deal, and even
19 to this day, I don't know a great deal, about how
20 they're made. And they include complex parts,
21 electronics, software, and so on.
22 So because of the speed with which we were going to
23 require machines to arrive into clinical practice, we
24 had to moderate some of the clinical ambition about what
25 these machines would be able to do, because otherwise we



1 would set a specification that was unachievable, and
2 patients would die as a result of not having machines
3 available to them.
4 So the specification that we wrote gave the basic
5 requirements of a machine that could keep a patient who
6 needed it alive for a few days, potentially, until
7 something better was available.
8 So the basic requirements included a machine that
9 would be able to support a patient who was fully
10 sedated, unable to make any respiratory effort
11 themselves, and an optional requirement for a machine
12 that would be able to support the breathing of a patient
13 who was making some respiratory effort.
14 Now we take both of those modes for granted in
15 normal practice but the second one of those is really
16 quite complicated because the machine could potentially
17 interfere with the patient's respiratory effort, and it
18 would not -- it would be a disaster. So we set the
19 minimum requirement as just the machine would do all the
20 breathing for the patient.
21 Similarly, the amount of oxygen that we give
22 patients -- normally, on any ventilator, we can dial up
23 at any percentage between 21%, which is room air, and
24 100%. But that requires a degree of engineering,
25 potentially software, that we considered would be



1 unachievable in the timeframes that we were working to.
2 So we set a minimum requirement of 40 to 50% oxygen or
3 90 to 100% oxygen, and that, we felt, was achievable and
4 also, as safe as it could be, and so on and so forth.
5 **Q.** Yes, and I'm sure there were many others. And some of
6 them were about, is this correct, trying to mitigate
7 clinician unfamiliarity, as well, were they, sort of
8 trying to make something universal?
9 **A.** Something that would be easily adopted into practice,
10 something very simple that would be failsafe, and
11 potentially deployed with clinical staff that were
12 unfamiliar with the critical care environment, as well
13 as with the machine.
14 **Q.** You were asked to produce this, you say, within one
15 working day on 13 March 2020?
16 **A.** That is correct.
17 **Q.** Which with some under-statement, you say, "This was duly
18 provided". So you did that very quickly. It was
19 something you say you did in -- you developed together
20 with your husband, Mr Pennington-Ridge, Edward
21 Pennington-Ridge; is that right?
22 **A.** That is correct.
23 **Q.** You say he's an independent design consultant who
24 fulfilled the role of "inventor in residence at UCLH"
25 and had extensive and relevant design experience.



1 **A.** That is correct. He is a designer that has specialised
2 over the years in technology solutions for low resource
3 settings, and that's pretty much the situation we
4 were in.
5 **Q.** One of the statements we've received, as you're aware,
6 is from Professor Tom Clutton-Brock, a name we've
7 mentioned a few times, director of the Medical Devices
8 Testing and Evaluation Centre with whom, ultimately, you
9 would work quite closely again, wouldn't you, on testing
10 the designs that you'd said, "yes" and "no" to?
11 Can we just turn that up, please, INQ000515729.
12 Can you see, halfway through paragraph 4,
13 Professor Clutton-Brock says:
14 "In 2020 other members of MD-TEC and the Trauma MIC
15 [which is the Trauma Management MedTech Co-operative]
16 did not have expertise in mechanical ventilation, they
17 assisted with phone calls and deliveries, but did not
18 take part in any ventilator testing during the
19 'Challenge'. To assist with administrative duties and
20 to act in training videos, my two daughters, history
21 undergraduates who had come home from university,
22 volunteered to come into MD-TEC each day with me."
23 Now, on the one hand you working with your husband,
24 Professor Clutton-Brock pulling in his undergraduate
25 daughters, again, it's that spirit of ingenuity and

1 resourcefulness in a crisis, on the one hand, isn't it?
2 **A.** Absolutely.
3 **Q.** But again, does this tell us -- the next clinical care
4 director may not have a husband or the next director of
5 MD-TEC may not have willing daughters to come and help
6 him in the crisis. What does this tell us? What
7 lessons can we learn, looking forward,
8 about (inaudible)?
9 **A.** My overarching lesson is that we went into the pandemic
10 really unprepared in terms of the technology that we had
11 available to us. We were really unprepared to be able
12 to scale up critical care capacity in the way that we
13 had to, and woefully unprepared for what we thought
14 might be our reasonable worst-case scenario. So in
15 order to avoid the luck, or in order to not need the
16 luck that we had through these sorts of -- in these
17 circumstances, it's all about planning, and
18 preparedness, and spending a year or even five years
19 doing what we had to do in a few days.
20 **Q.** To some extent a similar theme runs through the -- your
21 assembly of the Technical Design Authority. We've heard
22 a little bit about that. This was group of senior
23 clinicians you assembled between -- within a very short
24 space of time between 15 and 18 March for which you
25 provided clinical leadership and which served, is this



1 right, as the independent expert advisory panel to the
2 Cabinet Office in the Ventilator Challenge?
3 **A.** That is correct, and some of them also assisted with the
4 procurement effort of devices that were already CE
5 marked and were being procured from overseas, as
6 Mr Stirling has described this morning.
7 **Q.** You say this:
8 "... no formal appointments process was undertaken,
9 due to the speed with which we needed to assemble [that]
10 panel."
11 It was you, was it, asking who you thought could
12 help?
13 **A.** Yes, and there was a bit of rhyme and reason to it in
14 that I asked clinical colleagues who, first of all,
15 I trusted and I knew would have the required expertise.
16 I asked one colleague with a respiratory medicine
17 background as well as intensive care; another colleague
18 with an anaesthetics background as well as intensive
19 care. We have slightly different training and so
20 slightly different experience. Another anaesthetist
21 intensivist who had been involved in procurement
22 decisions in his own trust which I thought would also be
23 useful. And a senior, what we would call resident now,
24 who was already working with NHS England in other
25 matters, and also was able to provide a sort of hands



1 on, the residents are always there around the clock. So
2 that perspective as well.
3 And the other thing I had to take into consideration
4 when asking these colleagues to help was their
5 availability.
6 So I did actually ask others, obviously people in my
7 own hospital, as you would expect, but they were all too
8 busy either delivering clinical service or planning for
9 delivering clinical service to be able to give their
10 time to this.
11 So availability and expertise were the key
12 considerations.
13 **Q.** I'm going to take you forward to one of the reflections
14 you have, and the lessons learned, and it's your
15 paragraph 200(c), you say:
16 "In advance of any future emergency, a panel of
17 experts, including engineers and clinicians should be
18 commissioned to develop the minimum acceptable
19 specifications for a range of scenarios, based on the
20 demand and the timeframe for manufacture ..."
21 Which, you can say:
22 "... would improve the feasibility and effectiveness
23 of our response [in] a significant surge ..."
24 And we'll come to the reasons in a minute. But
25 would that avoid the need for husbands, daughters -- no



1 doubt qualified -- but somewhat hastily assembled
2 panels?
3 **A.** It would, and it would also give clinicians more trust
4 in what was going to be delivered to them. There was
5 a lot of an anxiety -- so there was a phrase that
6 I heard earlier on that I was troubled by, which is that
7 "a bad ventilator is worse than no ventilator at all".
8 Now, perhaps counter-intuitively, even if you need
9 mechanical ventilation, because you're effectively
10 suffocating because of your respiratory illness,
11 ventilators can cause further damage to the lungs. If
12 we put too much air and oxygen into you, you can cause
13 what we call volume trauma, you inflate the lungs too
14 much. If you apply too high a pressure you can cause
15 barotrauma, pressure damage to the lungs. So in normal
16 practice we are all trained to apply just the right
17 amount of volume and just the right amount of pressure
18 to the patient in order to try to optimise their oxygen
19 and carbon dioxide levels without causing further damage
20 to the lungs.
21 And we've done many randomised trials to try to find
22 the right sweet spot. The problem is that, in order to
23 get that sweet spot, you need the most complex, the most
24 sophisticated of intensive care ventilators. And we
25 were not in a position to be able to make, and indeed

1 even procure, a lot of the time, ventilators of that
2 sophistication.
3 Therefore, we had to make some difficult choices to
4 provide a pragmatic specification that would be
5 achievable by industry, and save some lives.
6 And so that statement of "a bad ventilator is worse
7 than no ventilator at all" was completely untrue, in the
8 complex of the pandemic, because there would have been
9 potentially, if the surge in demand had been as much as
10 we had feared it would be, patients who would have
11 simply been left to die without that support. So we had
12 to make those choices.
13 **Q.** Now, I'm not going to go into the detail of the
14 evolution, but it's fair to say, isn't it, that the
15 specification you provided developed?
16 **A.** Yes.
17 **Q.** It developed quite significantly over time?
18 **A.** It did. And that is because the circumstances changed.
19 So the specification that was initially published by the
20 MHRA on 18 March, based off that initial one that my
21 husband and I produced, was the most basic, anticipating
22 that we would have to have product, as Sir Gareth
23 mentioned, within two to three weeks.
24 Now, what happened was two things. The first is
25 that, thankfully, the catastrophic surge in demand that



1 we anticipated did not happen. And that was largely, if
2 not entirely, down to the non-pharmaceutical
3 interventions, the lockdowns.
4 So that bought us some time. And what that meant
5 was that we could then afford to push for a slightly
6 more sophisticated minimum specification, because the
7 people that were developing the ventilators had more
8 time to develop that degree of sophistication.
9 Then the second thing that happened was that our
10 understanding of Covid as a disease was improving,
11 because we were actually treating patients in their many
12 hundreds and thousands, and so we could identify the
13 specific things that it would be particularly important
14 to try to prioritise in a more sophisticated design.
15 So one of those, as Sir Gareth mentioned a couple of
16 weeks ago, was the ability to put a tube down
17 the breathing tube and suck out the phlegm, essentially,
18 because the patients were -- they had a lot of what we
19 call secretions on their chest.
20 That's a 'nice to have' on the context of 18 March,
21 but it became essential as we bought more time.
22 **Q.** Now, one of your roles, clinically, was to provide what
23 you said a 'go/no go' recommendation?
24 **A.** Yes.
25 **Q.** So you're presumably testing offers of procurement from



1 everywhere, overseas, the UK and through the Ventilator
2 Challenge, and testing them against this specification,
3 are you and then deciding "Yes, procure" or "No, don't
4 procure"?
5 **A.** So the procurement and the Ventilator Challenge required
6 slightly different approaches.
7 In procurement, we were looking at devices that all
8 had CE marks, and so they'd been through some sort of
9 regulatory approvals process, perhaps not in the UK, but
10 internationally, and my colleagues and I reviewed the
11 publicly available materials, so the instructions for
12 use from manufacturers and other materials from
13 manufacturers' websites, and, based off that
14 information, categorised the ventilators into proper ICU
15 intensive care ventilators versus anaesthetic machines
16 versus transport ventilators, in reducing levels of
17 sophistication, and then provided that 'go/no go', based
18 entirely on clinical grounds, to Chris Stirling and his
19 team.
20 **Q.** So your considerations were entirely clinical, but, from
21 what you say, this is your paragraph 97, that
22 information from manufacturers, you were entirely
23 dependent on that, given the circumstances of the
24 procurement?
25 **A.** Yes, if they were machines that were not already in use



1 in the NHS.
2 **Q.** Was the quality of that information provided by
3 manufacturers variable?
4 **A.** As it turns out, it was variable.
5 **Q.** You also say this: the clinical review could not involve
6 putting the devices to actual use, as you had presumably
7 to make such rapid procurement decisions in the context
8 of high international demand and shortage?
9 **A.** That is correct. And going to do back over my records,
10 I -- you can see the pace with which everybody was
11 working. The procurement team would send us huge
12 spreadsheets and sets of documents to review, and then,
13 in the nicest possible way, were pushing us very, very
14 hard for decisions because the market was so volatile
15 that we had to make those -- collectively had to make
16 those decisions quickly in order to not miss out on the
17 opportunity to buy the machines.
18 **Q.** We don't need to dwell on this in particular detail
19 because we've heard evidence, but did you hear some of
20 the evidence of Mr Stirling this morning?
21 **A.** I did.
22 **Q.** Including about the Shangrila model, the 510S?
23 **A.** I did.
24 **Q.** You will have heard then, I expect, the consternation
25 that caused to the particular trust to which it was

1 delivered?
2 **A.** Yes.
3 **Q.** You in fact received, didn't you, correspondence from
4 that trust directly to you?
5 **A.** I did.
6 **Q.** With those concerns?
7 **A.** At my request.
8 **Q.** At your request, right.
9 So if we look very briefly, please, INQ000508296.
10 This a letter from Dr Jon Hulme, dated 13 April 2020,
11 just the first couple of lines:
12 "Dear Ramani,
13 ... Feedback summary of [this particular model of]
14 ventilator.
15 "We spoke with you today as part of a group of
16 clinicians and senior managers within the Birmingham and
17 Black Country region about our serious concerns that the
18 250 Shangrila 510 ventilators delivered to us are not
19 fit for purpose within an intensive care setting."
20 Then this:
21 "We believe that if used, significant patient harm,
22 including death, is likely."
23 So these were really significant concerns, weren't
24 they?
25 **A.** They were, and I think this is an example of how our



1 systems were actually -- although this outcome was
2 obviously not desirable, this is an example of how our
3 systems were working well, in that this issue had been
4 brought to my attention the previous day by a regional
5 medical director for the Midlands region who had
6 expressed it at one of the daily meetings that we had
7 with medical directors that Mr Stirling described this
8 morning. I then requested a meeting with the clinical
9 trials so I could hear their concerns firsthand.
10 Having heard them, and taking them very seriously,
11 I asked them to write this letter, which they did,
12 summarising everything, and also summarising where
13 the -- their experience of using the machines differed
14 from what the expectation should have been from the
15 manufacturers' materials.
16 So I subsequently, having received this letter, sent
17 immediate notice to the procurement team and the team
18 dealing with ventilator allocation to say, "Please don't
19 allocate any more, please don't buy any more," and "We
20 need to do some more due diligence."
21 I then, the next day, went to the next of these
22 meetings and heard from colleagues in the south-east,
23 from Portsmouth, where they had also received one of
24 these machines. And we basically collated through this,
25 and from Portsmouth the evidence that we needed to



1 confirm that we should not be trying to deploy these
2 machines into clinical practice. And we then sent one
3 of these machines to MD-TEC, to Birmingham, to do bench
4 testing which then eventually led to our final
5 conclusion that we should not deploy these machines.
6 **Q.** Part of the statement of the BMA suggests that there was
7 a rush to procurement process. First of all, do you
8 agree with that characterisation?
9 **A.** "Rushed" feels pejorative. We worked at pace.
10 **Q.** Do you think it was -- a device like this where you
11 hadn't been able to undertake the due diligence was
12 a consequence of that pace, then?
13 **A.** If we had years -- so if I was to support my colleagues
14 at my hospital in buying a new intensive care ventilator
15 today, that process would take months. My colleague in
16 charge of procurement would seek the best models, the
17 manufacturers, they would come to the hospital, they
18 would give us one to play with, or a few, and then the
19 team would make a collective decision. There was
20 clearly no time for any of that.
21 So again, preparedness was one thing that might have
22 assisted in this.
23 But again, for further context, the programme that
24 Mr Stirling was in charge of, by the middle of 2021, had
25 deployed more than 45,000 capital pieces of equipment



1 into the NHS, of which about 11,000 were mechanical
2 ventilators, non-invasive ventilation machines, CPAP
3 machines, and so on, and others were things like
4 analysers for blood at the bedside and so on.
5 To my knowledge, this is the only procurement that
6 failed in this way. So I think, given that context,
7 yes, we worked at pace, but it was to meet a need that
8 actually was largely met.
9 **Q.** Final aspect of questions on this particular model. If
10 we look, please, at INQ000514240, this is the MD-TEC
11 report on this particular model and it's just these
12 concluding paragraphs. So this is by
13 Professor Clutton-Brock who says:
14 "The extraordinary circumstances facing the NHS and
15 many other healthcare systems in March 2020 made the
16 usual process of device procurement (selection, medical
17 engineering training and assessment, clinical training
18 and assessment, detailed procurement instructions)
19 difficult, if not impossible, to follow."
20 Referring to the particular model the Shangrila, he
21 says:
22 "As a result, hose incompatibility, oxygen sensors,
23 casing and breathing system choices were not properly
24 identified."
25 And I wanted to ask you about this next line:

1 "Even in pandemic times rapid external review of new
2 technology procured for the NHS should be undertaken to
3 identify suitability and training needs."
4 Do you have a view on that?
5 **A.** Yes. And, first of all, I agree entirely with
6 Professor Clutton-Brock and what he describes as the
7 process that we evolved towards that Mr Stirling has
8 described this morning, where we procured the technology
9 based on written materials, because there was no time to
10 do anything else, but partly as a result of the
11 Shangrila episode, we set up the system that meant that
12 in Donnington where the supplies were received, there
13 was then a process that was undertaken with clinical due
14 diligence and an engineering assessment, and potentially
15 bench testing either at MD-TEC or in Nottingham to test
16 these devices before they were deployed into the system.
17 So we adapted our processes to get nearer to what
18 Professor Clutton-Brock describes here, or to meet that
19 request.
20 **Q.** I want to turn to your reflections, please, and put
21 three short parts of your witness statement to you, some
22 of which you may already have heard.
23 First, paragraph 115. You say:
24 "Although the target number of machines was
25 achieved ..."



1 Presumably you're talking about through the
2 Ventilator Challenge and the other procurement routes.
3 "Although [that] target number of machines was
4 achieved, the quality of the machines and their
5 suitability for use in the NHS, to treat Covid-19 or
6 other conditions was variable."
7 **A.** That is evidently true.
8 **Q.** "There were also shortages of other ... items, including
9 some medicines and consumable items. These limitations
10 meant that clinicians were not always able to treat
11 patients with their first choice of interventions."
12 **A.** That is also true.
13 **Q.** That's important, is it, or --
14 **A.** It is, because familiarity is one of the -- so human
15 factors in healthcare is a significant thing, and you
16 are less likely to make errors if you're using equipment
17 and everything that you're familiar with. You're also
18 less likely to make errors if you're not stressed, you
19 are not over worked, you're not having to look after
20 twice as many patients as usual, and so on and so forth.
21 So clearly the circumstances of the pandemic meant that
22 all of those things that on a normal day like today we
23 would be able to deal with didn't happen. So all the
24 human factors aspects of care were challenging.
25 **Q.** Next this, in terms of the number and the allocation.



1 It's your paragraph 120:
2 "[You're] not aware of any specific incident where
3 a patient who needed a ventilator was not able to access
4 one."
5 Is that still correct, to the best of your
6 knowledge?
7 **A.** That is still correct.
8 **Q.** You do go on to say, however -- you make a further
9 observation, don't you: you can't be sure that
10 individual decisions weren't affected by this -- the
11 aspect of what was available.
12 Can you just explain what you mean by that.
13 **A.** Yes. Even on a normal day, capacity constraints in the
14 intensive care, and quite possibly in other aspects of
15 healthcare, do affect decision making. So, for example,
16 you have, in the morning, five patients who have been
17 reviewed by an anaesthetist and a surgeon before coming
18 in for surgery, and it's been considered that they
19 should go to intensive care after their operation for
20 the benefit of their outcome, but you've only got two
21 intensive care beds available.
22 So then the clinicians on the day have to make
23 a decision about which to do, which two patients to put
24 those beds and what to do with the other three. Do we
25 cancel their procedure or do we proceed and send them to



1 the normal ward? So that's an example of where capacity
2 affects decision making on a day-to-day basis.
3 Now, you take the specific circumstances of the
4 pandemic and it is possible that anticipated pressure or
5 real pressure may have changed subtly the way that
6 clinicians thought about whether or not to send patients
7 to intensive care or escalate their care.
8 And we have heard in Module 3 from a survey that the
9 experience of clinicians was that escalation decisions
10 varied according to the demand in the system.
11 **Q.** Finally this, in terms of your reflections. You talk
12 about the efforts made by colleagues across DHSC,
13 Cabinet Office and NHS England as being remarkable and
14 effective in many ways. But you say this at 187:
15 "I do not consider that there was a robust system in
16 place to ensure an adequate supply of key healthcare
17 equipment and supplies to the NHS ..."
18 What's the basis for your observation there?
19 **A.** I think that Mr Stirling, Dame Emily and others set up
20 systems and processes rapidly that were as effective as
21 they could be in the circumstances. But at the risk of
22 being boring, we were not well prepared. And ideally,
23 we shouldn't have had to have stood those systems up in
24 haste at the end of February or beginning of March 2020;
25 we would have had a plan that could have been executed

1 earlier.
2 Q. I want to turn now to what you say are the lessons to
3 learn. This is your paragraph 200.
4 Could you just talk us through some of the key
5 aspects starting with paragraph (a), about stockpiles.
6 A. So there are some items that it's relatively
7 straightforward to stockpile: PPE, some consumable
8 items. Ventilators and other capital equipment, it's
9 a much more difficult proposition, for multiple reasons.
10 The first is they're bulky, big items, so they take up
11 a lot of space, so storage is challenging.
12 Second, they actually don't stay in the box and --
13 you can't keep it in the box and then ten years later
14 open it and expect it to work properly.
15 Some of the components of ventilators, oxygen
16 sensors, and so on, have a life expectancy. And so if
17 they're not used, they won't work.
18 And then the third is that familiarity aspect.
19 Ideally, these sorts of equipment should be being used
20 in clinical practice so that the clinicians are familiar
21 with them, and we reduce the human factors risks.
22 That said, we could do more to have at least a bit
23 of a buffer in terms of stockpiling and supply. The
24 most obvious way to do that is actually just to expand
25 critical care capacity.



1 I said at the beginning we have fewer critical care
2 beds per head of population than comparable countries.
3 If we increase that, we would use it, because we would
4 send more patients to critical care after their
5 operations, for example, and we would then have
6 a bigger, sort of, buffer in the event of a pandemic
7 surge or another emergency.
8 We could also have reserve stocks in hospitals that
9 were stored and rotated through the attachments
10 a regular basis. There's a challenge, because one of
11 the things in almost every hospital that is always at
12 a premium is storage space, but it is surely -- if we
13 were able to do things like the Ventilator Challenge,
14 then we should be able to do things like work the
15 storage space out.
16 So I think that we can do more. I think today,
17 because I looked at the numbers this morning, we have
18 4,000 critical care beds available. We have about an
19 80% occupancy of those beds, and we have another
20 800 beds that we could open in the event of a surge, an
21 unexpected surge.
22 We could do a bit more than that and build a bit
23 more of a buffer, I think.
24 Q. Thank you. You talk about this, you say an unhelpful
25 narrative developed in the press and:



1 "... disagreement between professionals and
2 professional bodies which focused on the quality and
3 quantity of ventilators and other items which were
4 procured or manufactured to meet surge demands. This
5 could be avoided through better preparation and advance
6 planning and agreement on the potential risks and
7 benefits of different approaches with all relevant
8 stakeholders -- including the general public."
9 What do you mean by "disagreement between
10 professionals and professional bodies"? How widespread
11 was that and how deep? What was that?
12 A. So there were disagreements in discussions which related
13 back to that point I made earlier about a bad ventilator
14 being worse than no ventilator at all. That kind of
15 notion. And some of that then spilled into the press,
16 and that was pretty challenging, actually, given the
17 efforts that we were making, simply to -- even just save
18 one life by having an additional machine that could
19 fulfil a basic purpose.
20 So -- but I do understand why colleagues felt that
21 way, because that's the way that we've all been trained,
22 and this was an extraordinary situation that was -- none
23 of us were ready for.
24 So the future lesson is: be ready. Have these
25 conversations in peacetime, not when we're all rapidly



1 trying to change the way that we work in anticipation of
2 these huge surges in demand.
3 Q. I have asked you about your suggestion for a panel of
4 experts, including engineers and clinicians. You go on
5 in that paragraph to talk about a number of other
6 factors: regulators, hibernating contract. Do you want
7 to pick out any of those that you think are relevant to
8 this --
9 A. Yes, so for -- so we know that one of the great
10 successes of the pandemic in this country was the
11 research, was the RECOVERY Trial, was the ISARIC study,
12 which studied the epidemiology of Covid and so on. Part
13 of the reason why those endeavours were so successful is
14 because, in the case of the ISARIC study and REMAP-CAP,
15 which was an intensive care study, the protocols for the
16 research were already written. So when the researchers
17 got wind of the fact that the pandemic was coming, they
18 just activated those. The regulators were ready, and
19 the processes that would normally take six months took
20 six days, for example.
21 So if we tried to adopt a similar approach with
22 this, then we might be better prepared in the future.
23 We could have -- we could model different scenarios. So
24 we could look at the scenario that faced us in sort of
25 15, 16 March 2020 where we thought we might need 30,000

1 ventilators within two to three weeks, and then you
2 could also model a different scenario where you only
3 needed 5,000 ventilators in six weeks, and so on and so
4 forth.
5 And then you develop both the specification that
6 meets professional and, importantly, patient and public
7 approval, and then you take that to the regulator and
8 you say, "Okay, so this is what we think we need to do.
9 What's the process going to be for getting this
10 approved?"
11 Get the regulatory approval in advance, and then the
12 whole process can get going quicker. And then all of
13 the work that Sir Gareth and others did, approaching
14 manufacturers, finding all the bits to go in the
15 ventilator, that can all be done in advance. You could
16 potentially plan to pivot industry from what it was
17 normally doing, building Formula One cars or whatever,
18 over to this, if anticipated.
19 So all that this is talking about really, is taking
20 a more considered, measured, long-term approach to the
21 challenge that we faced and doing it over years rather
22 than days.
23 **Q.** Just finally, please, your paragraph 202, the lack of
24 manufacturing capability for medical devices,
25 consumables and medicines, was a key limitation of your



1 response in your view?
2 **A.** Yes.
3 **Q.** What do you say should be done to look forward to that?
4 **A.** So, for example, one of the things that we ran short of
5 in the first wave was the fluids that are required to
6 support kidney failure. So in intensive care we use
7 a type of dialysis called haemofiltration, I don't want
8 to get too technical, but essentially you wash the
9 patient's blood, and you replace electrolytes, so
10 potassium, sodium and so on.
11 Now, to do that you need various bits of kit but the
12 most essential is just big bags of clean fluid with some
13 potassium and some sodium, and so on. This is not
14 a difficult thing to make. But we didn't have that
15 capability in the UK. And as a result of that, and the
16 supply pressures from the two manufacturers that make
17 those fluids that serviced UK critical care, our
18 clinicians were required to adopt completely different
19 practice to how they managed kidney failure in intensive
20 care. They changed -- even down to using something
21 called peritoneal dialysis, which is a way of providing
22 support to kidney failure patients at home, which I
23 don't think we'd ever used in intensive care before, but
24 again the ingenuity of my colleagues, they adapted and
25 they did it.



1 But that was a really good example of something that
2 actually I couldn't quite understand why we couldn't do
3 it ourselves, we couldn't manufacture the fluids
4 ourselves.
5 Ventilators are right the other end of the spectrum
6 in terms of complexity. In fact, the two manufacturers
7 that, won, if you like, the Ventilator Challenge, were
8 Penlon who make anaesthetic machines based in the UK,
9 and Smiths Medical who also have a factory in the UK and
10 they make transport ventilators.
11 So we had that capability that was adapted to the
12 Ventilator Challenge, but we didn't have any intensive
13 care ventilation ventilator makers in the UK. So that's
14 a limitation.
15 **Q.** Having said "finally", I hope my Lady will permit me
16 a final "finally", but briefly if you will, just looking
17 at the time, clinician choice. Choice max, as
18 Mr Stirling said it. Given that you're here, can you
19 give us the clinical perspective from where you are on
20 that particular aspect of the debate?
21 **A.** I can see it from every angle. So Lord Carter did
22 a review in 2015 of the unwarranted -- what he perceived
23 to be the unwarranted variation in procurement. He
24 looked at orthopaedic surgery and the huge variation
25 there was in the types of prosthetics that were used to



1 replace a hip between different trusts and hospitals and
2 the huge variation in price. And from that came the
3 getting-it-right-first-time initiative, which is about
4 trying to rationalise that huge variation, initially in
5 procurement, but it's moved into other things as well.
6 So too much variation, or too much choice max, is
7 probably undesirable, both from a cost perspective, but
8 more importantly from a safety perspective, because I've
9 been in the same place for 15 years or something, but
10 our residents rotate, our nurses will increasingly
11 rotate. And so if they rotate through different places
12 and they have to get used to different equipment, then
13 there's clearly a safety challenge there.
14 The flipside of that, is the scenario that I just
15 described with the renal replacement therapy fluids,
16 because there's two manufacturers that make those, and
17 all the kit associated with renal replacement therapy in
18 intensive care, and because of the international demand
19 for those, we ran out or we were close to running out,
20 and we had to modify our practice.
21 So Mr Stirling had a sliding scale like this this
22 morning, and he said we're over here in terms of choice
23 max [indicating]. So I'd like to be somewhere a bit
24 closer to the middle. And I think that what's required
25 is giving the clinical choice but in a sort of more

1 organised way. So get clinicians together to make
2 decisions across different institutions. There may
3 be -- I'm not a procurement expert -- there may be some
4 financial benefit to that as well, and I would
5 anticipate that there would be a patient safety benefit.
6 **MR STOATE:** Thank you, my Lady. Those are my questions.
7 There are some --
8 **LADY HALLETT:** Thank you. Just a few more questions for
9 you, Professor.
10 Ms Morris, who is sitting there.
11 **Questions from MS MORRIS KC**
12 **MS MORRIS:** Thank you.
13 Good afternoon, Professor. I ask questions on
14 behalf of the Covid Bereaved Families for Justice, and
15 my first topic is going to be about the impact of the
16 procurement problems as you've described them overall,
17 and I think you're likely to agree with me about this,
18 but if I can just take you to a couple of paragraphs of
19 your statement.
20 You said at paragraph 171 of your statement that the
21 outcome of procurement of equipment supplies wasn't
22 sufficient to enable healthcare workers to meet the same
23 standards of care to patients treated during the surges
24 of the Covid-19 demand compared to non-surge or
25 pre-pandemic times.



1 Would you agree with that?
2 **A.** I would.
3 **Q.** You said that patients were treated with equipment or
4 medicines which were either not designed for that
5 purpose or didn't provide best practice, and were
6 treated by staff who were not necessarily fully trained
7 to do so. Again, would you agree with that?
8 **A.** I would. So on equipment, that was, for example, the
9 repurposing of anaesthetic machines or transport
10 ventilators to treat critically ill patients. In the
11 case of medicines, that was, you know, we -- you'd
12 mentioned propofol earlier, so that was in short supply
13 and so we had to moderate. And the staff, critically --
14 and my colleagues worked under unbelievable pressure and
15 we had to bring in people that were not used to working
16 in intensive care to support the intensive care effort
17 and it was extremely hard for them, and you've heard
18 other testimony across the modules on that.
19 **Q.** Yes, thank you. And thank you for those examples.
20 You've said that these risks would have been
21 cumulative, and potentially have had an impact on
22 patient safety.
23 **A.** Yes.
24 **Q.** Thank you. Would you agree that this impact on patient
25 safety was created to a large extent by the lack of



1 proper planning, the lack of preparedness, and the lack
2 of procurement supplies ahead of the pandemic?
3 **A.** I think that better planning and preparedness could have
4 avoided at least some of the challenges that we had. If
5 we had, for example, the surge in ventilator demand, if
6 that could have been met with equipment that people were
7 more familiar with, then yes, that would have helped.
8 **Q.** Thank you. I'm going to touch on familiarity again in
9 a moment, because at paragraph 173 you exhibited
10 a Coroner's Prevention of Future Deaths report?
11 **A.** Yes.
12 **Q.** For the East London Coroner in relation to two
13 gentlemen, Mr Kishorkumar Patel and Mr Kofi Aning, who
14 both died in April of 2021, in which the wrong filter
15 was used with a breathing system in their intensive care
16 ventilator at the Nightingale Hospital.
17 That coroner's report highlights that these two
18 incidents were reported to be part of a cluster of
19 incidents of a similar nature. And you'll recall from
20 that report that there was a number of issues. There
21 was the non-standardised colour-coding of the filters,
22 I think, different types of filters with different
23 names, and sort of optional and variable positionings of
24 the filters. I think the coroner's words were it
25 "resulted in a very confusing situation".



1 Is that an example you're giving there of where,
2 lack of availability of resources and demand in the
3 first wave, clinicians were using unfamiliar or less
4 than ideal equipment, medicines and consumables, as you
5 say in your statement?
6 **A.** I think it's actually a broader point than even that.
7 So these human factors, as I've said, are cumulative.
8 There's the James Reason Swiss cheese model of how these
9 things line up and cause really big problems.
10 With the filters, as an example, this is a problem
11 across lots of different consumable items, medicines and
12 so on, that -- things that do -- bits of kit that have
13 vastly different, or even slightly different, purposes
14 can look very similar.
15 So we have that problem with medicines. So you can
16 have a vial of two different drugs, one of them is
17 basically water and one of them is something that might
18 stop you breathing, and if you're in a rush and you look
19 at them, they've got the same colour on the bottle, and,
20 you know, mistakes can happen. It's a similar example
21 here with these filters, where one manufacturer makes
22 a filter that does this, red, and the other one makes it
23 green, and there's confusion.
24 So this is a wider point than just Covid pandemics
25 and so on. And it speaks a little bit to what

1 Mr Stirling said this morning about interoperability.
2 We need some basic standards for all of the equipment
3 that we use, and particularly the consumable items.
4 That means that if you pick up something that's green,
5 it always does the same thing.
6 **Q.** Yes. So the broader point that applies in normal times,
7 business-as-usual times, and then there's particular
8 need, isn't there, in surge times, pandemic times, when
9 the psychological burden that I think both yourself --
10 **A.** Yes.
11 **Q.** -- and Mr Stirling referred to is even more acute in
12 frontline workers who are maybe in that rush and there's
13 additional pressures to make those critical decisions.
14 So you would agree that that kind of lack of familiarity
15 in those issues are even more acute in pandemic times?
16 **A.** Absolutely. And you have staff that are not normally
17 working in those environments working in them -- under
18 supervision, but it's not ideal.
19 **Q.** Thank you. And I think just to complete the thought, as
20 it were, you say in paragraph 187 of your statement
21 that:
22 "These factors combined posed a potential risk to
23 patient safety ..."
24 And now, as the interim director of patient safety
25 at NHS England, this is still your view?



1 **A.** It is.
2 **MS MORRIS:** Thank you very much.
3 Thank you, my Lady.
4 **LADY HALLETT:** Thank you, Ms Morris.
5 Mr Thomas. Mr Thomas is over there.
6 **Questions from PROFESSOR THOMAS KC**
7 **PROFESSOR THOMAS:** Good afternoon, Professor, can you hear
8 me?
9 **A.** I can, thank you.
10 **Q.** Professor, as a little bit of context, I represent
11 FEMHO, the Federation of Ethnic Minority Healthcare
12 Organisations. Your insights, Professor, into the
13 procurement processes for ventilators and PPE during the
14 pandemic are invaluable, especially given the challenges
15 faced by ethnic minority healthcare workers in accessing
16 suitable equipment.
17 So the focus of my questions will be to understand
18 how considerations of diversity, particularly concerning
19 ethnic minorities, were integrated into the procurement
20 process and design of these critical resources. So
21 that's just to give you an indication as to where I'm
22 going.
23 So, at paragraph 54 to of your statement, you
24 mention that the approach to procurement of ventilators
25 had sufficient regard to equality and diversity



1 including the Public Sector Equality Duty ensuring that
2 different ventilators may be required for children than
3 adults, and you say this was factored into the
4 procurement decisions.
5 **A.** Yes.
6 **Q.** Could you elaborate, please, on how considerations for
7 diversity particularly concerning ethnic minority
8 healthcare workers and patients were integrated into the
9 ventilator procurement process, please.
10 **A.** Thank you, yes. So when it comes to the capital
11 equipment, like ventilators, non-invasive ventilation
12 machines and so on, there are no differences between the
13 types of machine that we would use in patients based on
14 their ethnic -- ethnicity. There are differences in the
15 types of machines that are suitable for patients of
16 different weights, predominantly, and therefore age.
17 So neonates -- people up to 3.5 kilograms, there's
18 a particular sort of category. Beyond that, from
19 3.5 kilograms basically up to any weight, most
20 ventilators would be able to cope with, although some of
21 the transport ventilators and the anaesthetic machine
22 ventilators that we had to repurpose for use in
23 critically ill patients would have been less able to
24 cope with very -- with patients of a very heavy weight.
25 There's a slightly different consideration in terms



1 of some of the consumable items, particularly facemasks,
2 and particularly the tight-fitting facemasks that we use
3 in patients requiring CPAP or non-invasive ventilation.
4 **Q.** And we've heard a lot often evidence about that.
5 **A.** Yes, and so the main consideration is facial hair, and
6 if you have facial hair, these masks do not fit as well.
7 The mask fit is not brilliant for all patients all of
8 the time at any time. I don't know that that's got
9 anything particularly to do with ethnicity. I think it
10 is some patients have different facial features.
11 A solution to that problem, potentially, is to use
12 an alternative what we call interface to provide
13 non-invasive ventilation and CPAP. So instead of a mask
14 that fits tightly on to the face, you can use a helmet
15 or a hood. And we tried to procure helmets and hoods at
16 high volumes as well as masks.
17 Apart from the facial hair and facial features
18 aspect, actually, some patients just prefer hoods and
19 helmets, because having a tight-fitting facemask on can
20 be really unpleasant. So we tried to procure
21 a diversity of interfaces.
22 What I don't know is whether the preferred interface
23 would have been available for every patient every time.
24 **Q.** All right. Let me move on to my next question, which is
25 what specific measures were taken to ensure that the

1 considerations in relation to the Public Sector Equality
2 Duty, that you reference in your statement, were upheld
3 when making those procurement decisions?
4 **A.** So, as I say, I think with -- with the capital
5 equipment, that -- I don't think that patients of
6 different ethnicities or, indeed, of other protected
7 characteristics would have been advantaged or
8 disadvantaged by particular devices in different ways.
9 When it comes to healthcare workers themselves, we
10 followed the approach that we do with normal
11 procurement, which is that we tried to deploy into the
12 system either equipment that there's a degree of
13 familiarity with, or provide training materials to
14 support that. And so I think Mr Stirling mentioned
15 quite exhaustive work that we did on developing
16 quick start guides, training videos, and all sorts of
17 things to go out into the system to support staff to use
18 these new bits of kit safely.
19 **Q.** Okay. I think you've touched on and answered my second
20 question, so I'm going to move on to my third.
21 You note the diversity of the NHS workforce and the
22 need for PPE design to accommodate all protected
23 characteristics, including ethnicity, as mentioned in --
24 that's paragraph 177 of your statement. So, in relation
25 to that, how were the diverse needs of the NHS and



1 social care staff considered in procurement in the
2 design of PPE during the pandemic?
3 **A.** I am afraid I can't answer that question, because PPE
4 was not an area that I was involved in at all.
5 **Q.** All right. I shall move on, then. Just bear with me.
6 **LADY HALLETT:** I think the rest are all PPE, aren't they,
7 Mr Thomas?
8 **PROFESSOR THOMAS:** My Lady, you're right. If you can't
9 assist on that, those are all my questions.
10 Thank you, my Lady.
11 **LADY HALLETT:** Thank you, Mr Thomas.
12 Those are all the questions we have for you,
13 Professor. Thank you so much for all you did during the
14 pandemic to try to assist your patients, and your
15 colleagues' patients, and thank you, also, to your
16 husband.
17 **THE WITNESS:** Thank you.
18 **MR STOATE:** Who obviously stepped up to when he was asked
19 to, or maybe you insisted that he did. Anyway, thank
20 you very much for your help.
21 **THE WITNESS:** Thank you.
22 **LADY HALLETT:** Thank you, Mr Stoate.
23 I shall return at 10.00 tomorrow morning.
24 **(4.09 pm)**
25 **(The hearing adjourned until 10.00 am the following day)**



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